

Fields, Flux, and the Big Theorems

Green, Stokes, Gauss. Three names, three theorems — and one idea, which you already met in Chapter 0.2.

WHERE WE ARE

Everything so far has been about functions: you feed in a point, you get out a number or a vector. Physics from Chapter 2.6 onward is not about functions in that sense. It is about **fields** — a number, or a vector, attached to *every point of space at once*. The electromagnetic field, the metric of spacetime, the Higgs field, every quantum field: all of them. So this chapter builds the calculus of fields, which means answering two questions. How much of a field is flowing out through a surface? And how much is circulating around a loop?

Those two questions produce the **divergence** and the **curl**, and then produce three theorems that every physics course states and almost none derives: Green's theorem, Stokes' theorem, and the divergence theorem. Here is the headline, and it is worth having before you start rather than after you finish.

They are not three theorems. They are one theorem, and it is the Fundamental Theorem of Calculus from Chapter 0.2 wearing different numbers of dimensions. Chapter 0.2 proved that $\int_a^b f' = f(b) - f(a)$ by chopping an interval into pieces and noticing that every interior endpoint appears twice with opposite signs and cancels, so only the boundary survives. Every theorem in this chapter is that argument, in a region of higher dimension, with faces or edges cancelling instead of points. By §5 you will have seen the whole family collapse into a single line, $\int_M d\omega = \oint_{\partial M} \omega$, which Chapter 3.5 will make precise and which is the most reused structural fact in theoretical physics.

Two sections carry more than their weight. §2 contains a genuine surprise: a field can be curl-free everywhere it is defined and *still* have no potential, because of a hole in the domain — and physics can measure the difference. That is topology entering physics, in Part 0. And §3 discovers that the divergence is nothing but the *trace of the Jacobian*, which by Chapter 0.4's identity makes it the fractional rate at which a flow changes volume. That one line turns Liouville's theorem (Chapter 1.3) into a triviality.

TOOLS YOU'LL NEED — Chapter 0.2: the integral as a limit of tagged sums, both halves of the Fundamental Theorem, and above all the telescoping argument that produced it. Chapter 0.4: the determinant as the factor by which a linear map scales volume, the trace, and the identity $\det(I + \epsilon A) = 1 + \epsilon \operatorname{tr} A + O(\epsilon^2)$. Chapter 0.5: the inner product, and the spectral theorem for symmetric matrices (used once, in §4). Chapter 0.6: the gradient, the Jacobian, the chain rule along a curve, Clairaut's theorem on mixed partials, and the change-of-variables formula. §3 and §4 each cash a specific theorem from Chapter 0.4; §7 is Clairaut twice.

1 · Scalar and vector fields

A **scalar field** on a region $U \subseteq \mathbb{R}^n$ is a function $\phi : U \rightarrow \mathbb{R}$: one number per point. Temperature through a room, pressure through the atmosphere, electrostatic potential, the density of a fluid, the concentration of a drug in tissue. A **vector field** is a function $\mathbf{F} : U \rightarrow \mathbb{R}^n$: one vector per point. Fluid velocity, the electric field, the gravitational field, the current density of a conserved substance.

We write the components of a vector field as functions of position,

$$\mathbf{F}(\mathbf{x}) = (F^1(\mathbf{x}), F^2(\mathbf{x}), \dots, F^n(\mathbf{x})), \quad (0.7.1)$$

with the index upstairs, following the convention of Chapter 0.6 §4: vector components carry upper indices, derivatives produce lower ones. In three dimensions we will freely write $\mathbf{F} = (F^x, F^y, F^z)$ when that is clearer.

1.1 · Field lines

The standard picture of a vector field is a set of **field lines**: curves that are everywhere tangent to $\mathbf{F}$. A field line through $\mathbf{x}_0$ is the solution of

$$\frac{d\mathbf{x}}{ds} = \mathbf{F}(\mathbf{x}(s)), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (0.7.2)$$

which is a system of ordinary differential equations — Chapter 0.8's subject. Two facts about those solutions are worth stating now because the pictures depend on them. If $\mathbf{F}$ is continuously differentiable then through every point there passes *exactly one* field line, so field lines never cross except where $\mathbf{F} = \mathbf{0}$. (This is the existence-and-uniqueness theorem for ODEs, which we quote here and Chapter 0.8 states properly.) And if $\mathbf{F}$ is a velocity field, the field lines are the paths that particles actually follow, so (0.7.2) is not a visualisation device — it is the equation of motion.

Field-line density is the usual proxy for field strength, and §5 will tell you exactly when that proxy is honest: field lines can only begin or end where the divergence is nonzero, which is why "lines of $\mathbf{E}$ start on positive charges and end on negative ones" is a theorem rather than a picture.

1.2 · A vector field is not a vector

This deserves saying once, plainly, because the notation actively hides it and Chapter 3.2 will charge you for the misunderstanding.

A single vector is one arrow. A vector field assigns an arrow *to each point*, and the arrow attached to $\mathbf{p}$ lives in its own private copy of $\mathbb{R}^n$ — the space of displacements *from* $\mathbf{p}$. Nothing in the definition lets you compare the arrow at $\mathbf{p}$ with the arrow at $\mathbf{q}$, because they are elements of two different vector spaces that merely happen to look identical.

In flat space with Cartesian coordinates they *are* identical, which is why you have never had to think about it: you slide arrows around freely, subtract $\mathbf{F}(\mathbf{q}) - \mathbf{F}(\mathbf{p})$, and get the derivative. That move is the one thing that stops working on a curved space. Take the sphere: a tangent vector at the north pole lies in a plane, a tangent vector at the equator lies in a different plane, and there is no basis-free way to say they point "the same way". The collection of all these private copies — one vector space glued to every point — is the **tangent bundle**, and it is the central object of Chapter 3.2.

Repairing the subtraction, so that "the derivative of a vector field" means something on a curved space, is the **covariant derivative** of Chapter 3.3, and the leftover from that repair is the gravitational field.

For this chapter we are in flat space with Cartesian coordinates and we will slide arrows around without apology. Just know which cheque is being written.

2 · Line integrals, conservative fields, and a topological surprise

2.1 · The definition

Start with the physics that forces the definition. A constant force $\mathbf{F}$ acting through a straight displacement $\mathbf{d}$ does work $W = \mathbf{F} \cdot \mathbf{d}$; the dot product is there because only the component of the force *along* the motion does anything, which is Chapter 0.5's orthogonal projection wearing a physical hat.

Now let the force vary and the path curve. Chop the path C into N short segments with endpoints $\mathbf{x}_0, \dots, \mathbf{x}_N$, let $\Delta \mathbf{x}_i = \mathbf{x}_i - \mathbf{x}_{i-1}$, and pick a sample point on each segment. Over a short enough segment the force is nearly constant and the segment is nearly straight, so

$$W \approx \sum_{i=1}^N \mathbf{F}(\mathbf{x}_{i-1}) \cdot \Delta \mathbf{x}_i. \quad (0.7.3)$$

That is a Riemann sum of exactly the kind Chapter 0.2 built the integral from, and we define the **line integral** as its limit as the segments shrink:

$$\int_C \mathbf{F} \cdot d\mathbf{r} \equiv \lim_{\max|\Delta \mathbf{x}_i| \rightarrow 0} \sum_{i=1}^N \mathbf{F}(\mathbf{x}_{i-1}) \cdot \Delta \mathbf{x}_i. \quad (0.7.4)$$

To compute one, parametrise. Let $\mathbf{x}(t)$, $t \in [a, b]$, trace out C . Then $\Delta \mathbf{x}_i \approx \dot{\mathbf{x}}(t_i) \Delta t_i$ by the linearization of Chapter 0.1, and (0.7.4) becomes an ordinary one-dimensional integral:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \dot{\mathbf{x}}(t) dt. \quad (0.7.5)$$

Everything from Chapter 0.2 now applies. Note the two things the definition does *not* depend on and one it does: it does not depend on which parametrisation you chose, nor on how you chopped; it *does* depend on the direction of travel, and reversing it flips the sign. Both claims are proved in the grind box.

▼ **Grind box — the line integral is parametrisation-independent, and orientation is the only thing that matters**

Let $\mathbf{x}(t)$, $t \in [a, b]$, and $\mathbf{y}(u)$, $u \in [c, d]$, be two parametrisations of the same curve, related by a reparametrisation $t = \tau(u)$ with τ differentiable, $\tau(c) = a$, $\tau(d) = b$ and $\tau' > 0$ (an increasing change of clock). Then $\mathbf{y}(u) = \mathbf{x}(\tau(u))$, and by the chain rule along a curve (Chapter 0.6 §5.1),

$$\dot{\mathbf{y}}(u) = \dot{\mathbf{x}}(\tau(u)) \tau'(u).$$

Substituting into (0.7.5) and using the substitution rule of Chapter 0.2 with $t = \tau(u)$, $dt = \tau'(u)du$:

$$\int_c^d \mathbf{F}(\mathbf{y}(u)) \cdot \dot{\mathbf{y}}(u) du = \int_c^d \mathbf{F}(\mathbf{x}(\tau)) \cdot \dot{\mathbf{x}}(\tau) \tau'(u) du = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \dot{\mathbf{x}}(t) dt.$$

Identical. The line integral is a property of the curve and the field, not of the clock you used. This is the same mechanism that made the ordinary integral independent of the partition: the τ' produced by the chain rule is exactly cancelled by the τ' produced by the substitution rule.

Orientation. Now take $\tau' < 0$ — running the curve backwards, so $\tau(c) = b$ and $\tau(d) = a$. The same computation goes through but the limits are swapped, and Chapter 0.2's convention $\int_b^a = -\int_a^b$ gives

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}.$$

So a line integral carries a sign that encodes a direction. That is not a nuisance: it is the entire mechanism of §5. When two adjacent cells share an edge, they traverse it in *opposite* directions, the two contributions cancel, and only the outer boundary survives. Orientation is what makes the cancellation happen, which is why every theorem in this chapter comes with a right-hand rule attached.

A cheap sanity check on (0.7.4): if $\mathbf{F}$ is everywhere perpendicular to the path, every term in the sum vanishes and the integral is zero. A magnetic force does no work for exactly this reason, since $q\mathbf{v} \times \mathbf{B}$ is perpendicular to $\mathbf{v}$ by construction.

2.2 · Gradient fields, and the Fundamental Theorem hiding in plain sight

Suppose $\mathbf{F}$ is the gradient of some scalar field: $\mathbf{F} = \nabla\phi$. Put that into (0.7.5) and watch what happens.

$$\begin{aligned} \int_C \nabla\phi \cdot d\mathbf{r} &= \int_a^b \nabla\phi(\mathbf{x}(t)) \cdot \dot{\mathbf{x}}(t) dt \\ &= \int_a^b \frac{d}{dt} \phi(\mathbf{x}(t)) dt = \phi(\mathbf{x}(b)) - \phi(\mathbf{x}(a)). \end{aligned} \tag{0.7.6}$$

The middle step is Chapter 0.6's chain rule along a curve, $\frac{d}{dt} \phi(\mathbf{x}(t)) = \nabla\phi \cdot \dot{\mathbf{x}}$; the last is the Fundamental Theorem of Calculus from Chapter 0.2. Two tools, one line, and the result is worth stopping on:

THE GRADIENT THEOREM

$$\int_C \nabla \phi \cdot d\mathbf{r} = \phi(B) - \phi(A)$$

where A and B are the endpoints of C . **The integral of a derivative over a curve is the original function evaluated on the boundary of that curve** — and the boundary of a curve is its two endpoints, the far one counting positively and the near one negatively.

That is the Fundamental Theorem again, with a one-dimensional domain sitting inside a three-dimensional space instead of on the real line. The structure is identical: derivative, integrated over a region, equals the undifferentiated thing on the boundary. Hold that sentence; §5 will repeat it twice more with the dimension raised.

2.3 · Four conditions that want to be equivalent

A field that is a gradient is called **conservative**, because if $\mathbf{F}$ is a force then $-\phi$ is a potential energy and the total energy is conserved. Four conditions on a field $\mathbf{F}$ defined on a region U compete for that name:

CONDITION	
(A)	$\mathbf{F} = \nabla \phi$ for some single-valued ϕ on U
(B)	$\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints of C
(C)	$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed loop C in U
(D)	$\nabla \times \mathbf{F} = \mathbf{0}$ everywhere in U

We prove $(A) \Rightarrow (B) \Leftrightarrow (C) \Rightarrow (A)$ now, and $(A) \Rightarrow (D)$ in §7. The remaining arrow, $(D) \Rightarrow (C)$, is the interesting one, and it is *false* without an extra hypothesis about the shape of U .

(A) $\Rightarrow$ (B). Done: (0.7.6) gives $\phi(B) - \phi(A)$, which mentions the path nowhere.

(B) $\Leftrightarrow$ (C). Suppose (B). Any closed loop can be cut at two points into two paths C_1 and C_2 running from A to B ; the loop is C_1 followed by C_2 reversed, so by the orientation rule of the grind box its integral is $\int_{C_1} - \int_{C_2} = 0$ by (B). Conversely, suppose (C) and let C_1, C_2 be any two paths from A to B . Then C_1 followed by reversed C_2 is a closed loop, so $\int_{C_1} - \int_{C_2} = 0$. That is (B).

(C) $\Rightarrow$ (A). This is the one that builds something. Assume (C), fix a basepoint $\mathbf{x}_0 \in U$, and *define*

$$\phi(\mathbf{x}) \equiv \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{r}, \tag{0.7.7}$$

the integral taken along any path in U from $\mathbf{x}_0$ to $\mathbf{x}$. By (C), and hence (B), the answer does not depend on which path, so ϕ is a genuine single-valued function. Now compute $\partial_i \phi$ at $\mathbf{x}$. Choose the path that reaches $\mathbf{x} + t\mathbf{e}_i$ by first going to

$\mathbf{x}$ any way you like and then travelling along the straight segment $\mathbf{x} + s\mathbf{e}_i$, $s \in [0, t]$. On that segment $\dot{\mathbf{x}} = \mathbf{e}_i$, so $\mathbf{F} \cdot \dot{\mathbf{x}} = F^i$, and

$$\phi(\mathbf{x} + t\mathbf{e}_i) - \phi(\mathbf{x}) = \int_0^t F^i(\mathbf{x} + s\mathbf{e}_i) \, ds. \quad (0.7.8)$$

Divide by t and let $t \rightarrow 0$. The right-hand side is exactly the object the *first* half of the Fundamental Theorem (Chapter 0.2) differentiates, and it gives $F^i(\mathbf{x})$ provided F^i is continuous. Hence $\partial_i \phi = F^i$ for every i , i.e. $\nabla \phi = \mathbf{F}$. ■

So (A), (B) and (C) are the same condition wearing three costumes, and none of the three is checkable in practice: you cannot test infinitely many loops. Condition (D) is different — it is a *local* test, three derivatives evaluated at a point, and you can check it in ten seconds. Which is exactly why the failure of (D) $\Rightarrow$ (C) matters so much.

2.4 • The counterexample, in full

Take the plane with the origin deleted, $U = \mathbb{R}^2 \setminus \{\mathbf{0}\}$, and on it the field

$$\mathbf{F}(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right). \quad (0.7.9)$$

It is perfectly smooth on U ; the only bad point has been removed. Compute its curl — in the plane, the single surviving component $\partial_x F^y - \partial_y F^x$, derived in §4. By the quotient rule,

$$\begin{aligned} \partial_x F^y &= \partial_x \frac{x}{x^2 + y^2} = \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \\ \partial_y F^x &= \partial_y \frac{-y}{x^2 + y^2} = \frac{-(x^2 + y^2) + y(2y)}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}. \end{aligned} \quad (0.7.10)$$

They are equal, so $\nabla \times \mathbf{F} = \mathbf{0}$ at every point of U . Condition (D) holds everywhere, with no exceptions and no fine print.

Now integrate around the circle of radius R centred at the origin, parametrised by $\mathbf{x}(t) = (R \cos t, R \sin t)$ for $t \in [0, 2\pi]$. On that circle $x^2 + y^2 = R^2$, so

$$\mathbf{F} = \frac{1}{R}(-\sin t, \cos t), \quad \dot{\mathbf{x}} = R(-\sin t, \cos t), \quad (0.7.11)$$

and their dot product is $\sin^2 t + \cos^2 t = 1$ — the factors of R cancel exactly. Therefore

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} 1 \, dt = 2\pi, \quad (0.7.12)$$

for *every* radius R . Condition (C) fails, spectacularly and by a fixed amount. A curl-free field with a nonzero circulation.

Where did the potential go? It exists locally: on the right half-plane $x > 0$, the function $\phi = \arctan(y/x)$ has $\nabla \phi = \mathbf{F}$ (differentiate it and see). Geometrically ϕ is the polar angle θ , and Chapter 0.6's polar grind box already told us $\nabla \theta = \hat{\boldsymbol{\theta}}/r$, which in Cartesian components is $(-y, x)/r^2$ — precisely (0.7.9). So the "potential" of this field is the angle. And the angle is not a function on U : go once round the origin and it has increased by 2π . It is **multivalued**, and (0.7.12) is measuring exactly that increment.

The repair to the theorem is a condition on the *shape* of U , not on $\mathbf{F}$. A region is **simply connected** if every closed loop in it can be shrunk continuously to a point without leaving the region. A disc is; a disc with a puncture is not, because a loop around the puncture has nothing to shrink through. With that hypothesis the missing arrow works, and §5 supplies the proof in one line: if the loop C bounds a surface S lying inside U , Stokes' theorem gives

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A} = 0. \quad (0.7.13)$$

For the punctured plane there is no such S : any surface spanning a loop around the origin must contain the origin, where $\mathbf{F}$ does not exist. The theorem does not fail; its hypothesis does.

2.5 · Why a physicist should care: the potential knows more than the field

It is tempting to file §2.4 as a curiosity about a contrived function. It is not. It is the reason one of the central objects of modern physics — the gauge potential — is physically real.

Consider an infinite solenoid along the z -axis carrying a current. Inside, there is a uniform magnetic field $\mathbf{B} = B\hat{\mathbf{z}}$. Outside, $\mathbf{B} = \mathbf{0}$ exactly: the field is confined. Now §7 will show that $\mathbf{B}$ can be written as $\nabla \times \mathbf{A}$ for a **vector potential** $\mathbf{A}$, and §5's Stokes theorem gives, for any loop C encircling the solenoid,

$$\oint_C \mathbf{A} \cdot d\mathbf{r} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{A} = \int_S \mathbf{B} \cdot d\mathbf{A} = \Phi, \quad (0.7.14)$$

the total magnetic flux threading the solenoid. So outside the solenoid $\mathbf{A}$ is curl-free — its curl is $\mathbf{B}$, which vanishes there — and yet its circulation around the solenoid is $\Phi \neq 0$. That is (0.7.9) again, with the solenoid playing the part of the deleted origin. The region outside a solenoid is not simply connected.

Classically this is harmless: a charged particle outside feels force $q\mathbf{v} \times \mathbf{B} = \mathbf{0}$ and nothing happens. Quantum mechanically it is not harmless. Chapter 6.3 will derive that a charged particle's wavefunction, carried along a path, acquires a phase

$$\Delta\varphi = \frac{q}{\hbar} \int_C \mathbf{A} \cdot d\mathbf{r}, \quad (0.7.15)$$

so a beam split around either side of the solenoid and recombined shows an interference shift of $q\Phi/\hbar$ — determined entirely by the circulation of $\mathbf{A}$, in a region where $\mathbf{B}$ is identically zero. This is the **Aharonov–Bohm effect**, predicted in 1959 and measured (Chambers 1960; conclusively by Tonomura and co-workers in 1986 with a superconductor-shielded toroidal magnet, which closed the last loopholes about leakage fields). Those are experimental facts, quoted; this book will derive the phase in Chapter 6.3 but cannot derive the measurement.

The moral is exactly §2.4's:

WHAT THE COUNTEREXAMPLE IS TELLING YOU

The potential can carry physical information that the field strength does not. $\mathbf{B}$ is zero throughout the region the particle visits; $\oint \mathbf{A} \cdot d\mathbf{r}$ is not; the interference pattern moves. So the gauge potential A^μ of Chapters 2.6 and 6.3 is not a computational convenience that could in principle be eliminated in favour of the "real" fields. It is closer to the fundamental object, and the field strength is what you get by differentiating it.

Note also *which* quantity is physical. Not $\mathbf{A}$ itself — you may add $\nabla\chi$ to it without changing $\mathbf{B}$ (§7), and that changes $\int_C \mathbf{A} \cdot d\mathbf{r}$ for an open path. What is physical is the integral around a *closed* loop, which by the gradient theorem (0.7.6) is unchanged, since $\oint \nabla\chi \cdot d\mathbf{r} = 0$. The observable is the holonomy of the potential, and the reason it is observable is topological.

3 · Divergence, derived — and its identity with trace

3.1 · Flux

Fix a surface S with a chosen side, and let $\hat{\mathbf{n}}$ be the unit normal pointing out of that side. Write $d\mathbf{A} = \hat{\mathbf{n}} dA$ for the vector area element. The **flux** of $\mathbf{F}$ through S is

$$\Phi_S[\mathbf{F}] \equiv \int_S \mathbf{F} \cdot d\mathbf{A} = \int_S \mathbf{F} \cdot \hat{\mathbf{n}} dA. \quad (0.7.16)$$

The dot product is not decoration. Suppose $\mathbf{F} = \rho\mathbf{v}$, the mass per unit volume times the velocity of a fluid. In a short time dt , the material that crosses a small patch of area dA is the material that was sitting in an oblique cylinder with base dA and slant $\mathbf{v} dt$. The volume of such a cylinder is base times *perpendicular* height, which is $(\mathbf{v} \cdot \hat{\mathbf{n}}) dt dA$; anything moving parallel to the surface never crosses it. So the mass crossing per unit time is $\rho \mathbf{v} \cdot \hat{\mathbf{n}} dA = \mathbf{F} \cdot d\mathbf{A}$, and

WHAT FLUX MEANS

$\int_S \mathbf{F} \cdot d\mathbf{A}$ is the **rate at which stuff crosses** S , counted positive in the direction of $\hat{\mathbf{n}}$. For a closed surface with $\hat{\mathbf{n}}$ pointing outward, $\oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}$ is the net rate at which stuff leaves V .

3.2 · The definition, before any coordinates

Now shrink the region. If stuff is leaving a small region faster than it enters, something inside is producing it. Define the **divergence** of $\mathbf{F}$ at a point $\mathbf{p}$ as the outflow per unit volume in the limit of a vanishing region containing $\mathbf{p}$:

$$\boxed{\nabla \cdot \mathbf{F}(\mathbf{p}) \equiv \lim_{V \rightarrow 0} \frac{1}{|V|} \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}.} \quad (0.7.17)$$

Look at what that definition does *not* contain: no coordinates, no basis, no components. It is built from a flux and a volume, both of which are geometric. That matters, because the formula we are about to derive looks entirely coordinate-dependent, and it will be reassuring to know it cannot be.

3.3 · The box, done properly

Take the region to be a rectangular box centred at $\mathbf{p} = (x, y, z)$ with sides $\Delta x, \Delta y, \Delta z$, so $|V| = \Delta x \Delta y \Delta z$. It has six faces, which pair up.

The pair perpendicular to x . On the face at $x + \frac{\Delta x}{2}$ the outward normal is $+\hat{\mathbf{x}}$, so the integrand $\mathbf{F} \cdot \hat{\mathbf{n}}$ is F^x . On the face at $x - \frac{\Delta x}{2}$ the outward normal is $-\hat{\mathbf{x}}$ and the integrand is $-F^x$. Each face has area $\Delta y \Delta z$, and over such a small face F^x equals its value at the face's centre plus a correction that integrates away (the grind box does this bookkeeping exactly). So the two faces together contribute

$$\left[F^x\left(x + \frac{\Delta x}{2}, y, z\right) - F^x\left(x - \frac{\Delta x}{2}, y, z\right) \right] \Delta y \Delta z. \quad (0.7.18)$$

The bracket is a difference of the same function at two nearby points, which is precisely what Chapter 0.1 taught us to expand. Writing $F^x(x \pm \frac{\Delta x}{2}) = F^x(x) \pm \frac{\Delta x}{2} \partial_x F^x + O(\Delta x^2)$ and subtracting, the values at x cancel and the two first-order terms add:

$$\left[\dots \right] = \partial_x F^x \Delta x + O(\Delta x^3), \quad (0.7.19)$$

where the $O(\Delta x^2)$ terms also cancel by the symmetry of the two-sided difference, leaving $O(\Delta x^3)$. So the x -pair contributes $\partial_x F^x \Delta x \Delta y \Delta z = \partial_x F^x |V|$, plus something smaller.

The other two pairs. Identical arguments, with x replaced by y and by z . Adding all three pairs and dividing by $|V|$:

$$\frac{1}{|V|} \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A} = \partial_x F^x + \partial_y F^y + \partial_z F^z + O(\Delta). \quad (0.7.20)$$

Now take the limit and the correction dies:

$$\boxed{\nabla \cdot \mathbf{F} = \partial_x F^x + \partial_y F^y + \partial_z F^z = \partial_i F^i.} \quad (0.7.21)$$

The last form uses the summation convention of Chapter 0.6 §5.2, and note that the index structure is correct: one up, one down, summed. That is the formula every textbook writes down first. Here it is a *consequence*, and the definition it came from — (0.7.17) — is the one that survives a change of coordinates, a curved space, and Chapter 3.3.

▼ Grind box — the error terms in the box argument, honestly

Two steps above were waved at. Both are repairable, and it is worth seeing how, because the same two gaps reappear in §5.

(i) The field varies across each face. We replaced $\int_{\text{face}} F^x \, dA$ by $F^x(\text{face centre}) \times \text{area}$. Expand F^x about the face centre in the two in-face directions, writing the offset as (η, ζ) :

$$F^x = F_{\text{centre}}^x + \eta \partial_y F^x + \zeta \partial_z F^x + O(\Delta^2).$$

Integrate over the face, which is symmetric about its centre: $\int \eta \, dA = 0$ and $\int \zeta \, dA = 0$, so the two linear terms vanish *exactly*, not approximately. What remains is

$$\int_{\text{face}} F^x \, dA = F_{\text{centre}}^x \Delta y \Delta z + O(\Delta^4),$$

since the quadratic remainder is $O(\Delta^2)$ times an area $O(\Delta^2)$. Divided by $|V| = O(\Delta^3)$ this is $O(\Delta)$ and vanishes in the limit. The symmetry of the box did the work; had we used a lopsided region we would have had to keep the linear terms.

(ii) The limit must not depend on the shape of the shrinking region. (0.7.17) says " $V \rightarrow 0$ " without specifying how, but we computed with boxes. For a continuously differentiable $\mathbf{F}$ the answer is the same for any reasonable family of shrinking regions, and here is the mechanism: near $\mathbf{p}$ the field is $\mathbf{F}(\mathbf{p}) + J(\mathbf{p}) \mathbf{h} + o(|\mathbf{h}|)$ by Chapter 0.6's total derivative. The constant part $\mathbf{F}(\mathbf{p})$ contributes zero flux through *any* closed surface (as much goes in one side as out the other — or formally, apply §5 to a constant field). The linear part contributes exactly $\text{tr } J \cdot |V|$ for any region, which is §3.4. And the $o(|\mathbf{h}|)$ part contributes $o(|\mathbf{h}|) \times \text{area} = o(|V|)$ for any region whose surface area does not blow up relative to its volume. So the limit is $\text{tr } J$ regardless of shape.

That last observation is worth extracting: **the divergence only depends on the linear part of the field**, i.e. on the Jacobian. Which is the next subsection's entire content.

3.4 · Divergence is the trace of the Jacobian

Compare (0.7.21) with the Jacobian matrix of Chapter 0.6 §2.2, whose entries are $J_j^i = \partial_j F^i$:

$$J = \begin{pmatrix} \partial_x F^x & \partial_y F^x & \partial_z F^x \\ \partial_x F^y & \partial_y F^y & \partial_z F^y \\ \partial_x F^z & \partial_y F^z & \partial_z F^z \end{pmatrix}, \quad \text{tr } J = \partial_i F^i. \quad (0.7.22)$$

The divergence is *the trace of the Jacobian*. That is not a curiosity; combined with Chapter 0.4 it converts the definition into a statement about volume, and it does so in four lines.

Let $\mathbf{F}$ be a velocity field, so that a particle at $\mathbf{x}$ moves to $\mathbf{x} + \epsilon \mathbf{F}(\mathbf{x})$ in a short time ϵ . Call that map Φ_ϵ . Its derivative is

$$D\Phi_\epsilon(\mathbf{x}) = I + \epsilon J(\mathbf{x}), \quad (0.7.23)$$

differentiating $\mathbf{x} \mapsto \mathbf{x} + \epsilon \mathbf{F}(\mathbf{x})$ term by term. Chapter 0.6 §8 proved that a differentiable map multiplies the volume of a small region by $|\det|$ of its derivative. And Chapter 0.4 §6 proved the identity we now need:

$$\det(I + \epsilon A) = 1 + \epsilon \operatorname{tr} A + O(\epsilon^2). \quad (0.7.24)$$

Put the three together. A blob of volume $V(0)$ carried by the flow for a time ϵ has volume

$$V(\epsilon) = \det(I + \epsilon J) V(0) = \left(1 + \epsilon \operatorname{tr} J + O(\epsilon^2)\right) V(0), \quad (0.7.25)$$

so that, rearranging and letting $\epsilon \rightarrow 0$,

$$\boxed{\frac{1}{V} \frac{dV}{dt} = \operatorname{tr} J = \nabla \cdot \mathbf{F}.} \quad (0.7.26)$$

THE READING TO CARRY FORWARD

The divergence of a velocity field is the fractional rate of change of the volume of a blob carried along by the flow. Positive divergence: the blob expands. Negative: it contracts. Zero: the flow is *incompressible*, and "divergence-free" and "volume-preserving" are the same sentence.

And the reason is Chapter 0.4's one-line identity. The determinant is what a linear map does to volume; the trace is the *rate* at which a map changes volume when it barely does anything at all; and the Jacobian is the linear map that a flow is, locally, over a short time. Divergence is the composition of those three facts, and nothing else.

Integrating (0.7.26) along a trajectory (it is $\frac{d}{dt} \ln V = \nabla \cdot \mathbf{F}$, an equation of the type Chapter 0.1 solved) gives the finite version,

$$V(t) = V(0) \exp\left(\int_0^t \nabla \cdot \mathbf{F} \, dt'\right), \quad (0.7.27)$$

with the integral taken along the path of the blob. The figure below measures both sides of this.

One immediate dividend, banked now and spent in Chapter 1.3. A mechanical system's state is a point (q, p) in phase space, moving with velocity $\dot{q} = \frac{\partial H}{\partial p}$, $\dot{p} = -\frac{\partial H}{\partial q}$. That is a vector field on phase space, and its divergence is

$$\frac{\partial}{\partial q} \left(\frac{\partial H}{\partial p} \right) + \frac{\partial}{\partial p} \left(-\frac{\partial H}{\partial q} \right) = \frac{\partial^2 H}{\partial q \partial p} - \frac{\partial^2 H}{\partial p \partial q} = 0 \quad (0.7.28)$$

by Clairaut's theorem (Chapter 0.6 §6.1). Zero divergence, so by (0.7.26) phase-space volume is exactly conserved by Hamiltonian flow. That is **Liouville's theorem**, one of the deepest facts in classical mechanics and the foundation of statistical mechanics, and it has just cost us two lines.

▼ Grind box — divergence in spherical coordinates, from the flux definition

Because (0.7.17) mentions no coordinates, we can evaluate it in any we like. Take a "box" bounded by the coordinate surfaces r and $r + dr$, θ and $\theta + d\theta$, ϕ and $\phi + d\phi$. From Chapter 0.6 §8.2 its volume is

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi,$$

and the three pairs of faces have areas $r^2 \sin \theta \, d\theta \, d\phi$ (the r -faces), $r \sin \theta \, dr \, d\phi$ (the θ -faces) and $r \, dr \, d\theta$ (the ϕ -faces). Write the field in the orthonormal basis, $\mathbf{F} = F^r \hat{\mathbf{r}} + F^\theta \hat{\boldsymbol{\theta}} + F^\phi \hat{\boldsymbol{\phi}}$.

The r -pair. The outward normals are $\pm \hat{\mathbf{r}}$, so the net contribution is the difference of $F^r \times (\text{area})$ at $r + dr$ and at r — and the area itself depends on r , which is the entire novelty. So the difference is of the *product*:

$$\partial_r (F^r r^2 \sin \theta) \, dr \, d\theta \, d\phi.$$

Divide by dV : the $\sin \theta$ cancels and we get $\frac{1}{r^2} \partial_r (r^2 F^r)$.

The θ -pair. Same move; the area $r \sin \theta \, dr \, d\phi$ depends on θ through $\sin \theta$, so the net contribution is $\partial_\theta (F^\theta r \sin \theta) \, d\theta \, dr \, d\phi$, and dividing by dV gives $\frac{1}{r \sin \theta} \partial_\theta (\sin \theta F^\theta)$.

The ϕ -pair. The area $r \, dr \, d\theta$ does not depend on ϕ , so we get $\partial_\phi (F^\phi r) \, d\phi \, dr \, d\theta$, and dividing by dV gives $\frac{1}{r \sin \theta} \partial_\phi F^\phi$. Adding:

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F^r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F^\theta) + \frac{1}{r \sin \theta} \frac{\partial F^\phi}{\partial \phi}.$$

Every factor in that intimidating formula has now been accounted for: each one is an area or a volume from Chapter 0.6's Jacobian, and the derivatives act on the products because the areas themselves vary. Nothing was memorised.

The one that matters. For a purely radial field $\mathbf{F} = f(r) \hat{\mathbf{r}}$ only the first term survives, so $\nabla \cdot \mathbf{F} = r^{-2} (r^2 f)'$. Take $f = 1/r^2$:

$$\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} = \frac{1}{r^2} \frac{d}{dr} (r^2 \cdot r^{-2}) = \frac{1}{r^2} \frac{d}{dr} (1) = 0 \quad (r \neq 0).$$

The inverse-square field is divergence-free everywhere it is defined. Remember that; the warning callout below is about the four words in brackets.

A cross-check worth making. Chapter 3.3 will prove the general formula $\nabla \cdot \mathbf{F} = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} F^i)$, where $\sqrt{g}$ is the volume factor of Chapter 0.6 §8.3 and the F^i are *coordinate-basis* components. We quote it here only to check it against what we just derived. In spherical coordinates $\sqrt{g} = r^2 \sin \theta$, and the coordinate components are related to the orthonormal ones by $F_{\text{coord}}^\theta = F^\theta / r$ and $F_{\text{coord}}^\phi = F^\phi / (r \sin \theta)$, because ∂_θ has length r and ∂_ϕ has length $r \sin \theta$. Substituting reproduces the three terms above exactly. The flux box and the metric formula are the same statement.

4 · Curl, derived — and what a Jacobian is made of

4.1 · The definition

Divergence measured outflow per unit volume. The other thing a field can do near a point is go round, so measure circulation per unit area. Fix a direction $\hat{\mathbf{n}}$, take a small loop lying in the plane perpendicular to $\hat{\mathbf{n}}$ and traversed counterclockwise as seen from the tip of $\hat{\mathbf{n}}$ (the right-hand rule, which is a choice we are making and must then keep), and define

$$\hat{\mathbf{n}} \cdot (\nabla \times \mathbf{F})(\mathbf{p}) \equiv \lim_{A \rightarrow 0} \frac{1}{|A|} \oint_{\partial A} \mathbf{F} \cdot d\mathbf{r}. \quad (0.7.29)$$

Again: no coordinates. What we are asserting by writing the left side that way is that the resulting numbers, one for each direction $\hat{\mathbf{n}}$, are the components of a single vector — i.e. that the answer depends *linearly* on $\hat{\mathbf{n}}$. That is not obvious from (0.7.29), and the computation below establishes it, because the answer it produces is manifestly a dot product of $\hat{\mathbf{n}}$ with a fixed vector.

4.2 · The rectangle

Take $\hat{\mathbf{n}} = \hat{\mathbf{z}}$ and a rectangle in the plane $z = \text{const}$, centred at $\mathbf{p} = (x, y, z)$, with sides Δx and Δy , traversed counterclockwise. Walk the four sides, keeping track of direction:

SIDE	DIRECTION	CONTRIBUTION
bottom, at $y - \frac{\Delta y}{2}$	$+\hat{\mathbf{x}}$	$+F^x(x, y - \frac{\Delta y}{2}) \Delta x$
right, at $x + \frac{\Delta x}{2}$	$+\hat{\mathbf{y}}$	$+F^y(x + \frac{\Delta x}{2}, y) \Delta y$
top, at $y + \frac{\Delta y}{2}$	$-\hat{\mathbf{x}}$	$-F^x(x, y + \frac{\Delta y}{2}) \Delta x$
left, at $x - \frac{\Delta x}{2}$	$-\hat{\mathbf{y}}$	$-F^y(x - \frac{\Delta x}{2}, y) \Delta y$

Group the two y -directed sides together and the two x -directed sides together:

$$\oint \mathbf{F} \cdot d\mathbf{r} = \left[F^y \left(x + \frac{\Delta x}{2} \right) - F^y \left(x - \frac{\Delta x}{2} \right) \right] \Delta y - \left[F^x \left(y + \frac{\Delta y}{2} \right) - F^x \left(y - \frac{\Delta y}{2} \right) \right] \Delta x. \quad (0.7.30)$$

Each bracket is the same two-sided difference that appeared in (0.7.19), so each is a derivative times the spacing, up to higher order:

$$\oint \mathbf{F} \cdot d\mathbf{r} = \left(\partial_x F^y - \partial_y F^x \right) \Delta x \Delta y + O(\Delta^3). \quad (0.7.31)$$

Divide by the area $\Delta x \Delta y$ and let it shrink:

$$(\nabla \times \mathbf{F})_z = \partial_x F^y - \partial_y F^x. \quad (0.7.32)$$

The other two components come from the same computation with the labels rotated $x \rightarrow y \rightarrow z \rightarrow x$, since nothing in the argument singled out z :

$$\nabla \times \mathbf{F} = \left(\partial_y F^z - \partial_z F^y, \partial_z F^x - \partial_x F^z, \partial_x F^y - \partial_y F^x \right). \quad (0.7.33)$$

Each component is manifestly linear in $\mathbf{F}$ and the three of them assemble into a vector, which retroactively justifies the notation of (0.7.29). In two dimensions only the last component exists, and it is a *scalar*; that is the object §2 was computing.

4.3 · Every Jacobian splits in exactly one way

Now the structural fact, which explains what the curl *is* rather than how to compute it. Any square matrix J can be written as a symmetric part plus an antisymmetric part:

$$J = \underbrace{\frac{1}{2}(J + J^T)}_{S, \ S^T=S} + \underbrace{\frac{1}{2}(J - J^T)}_{A, \ A^T=-A}. \quad (0.7.34)$$

The split exists (just add the two pieces and the transposes cancel) and it is unique: if $J = S' + A'$ with S' symmetric and A' antisymmetric, transposing gives $J^T = S' - A'$, and adding and subtracting the two equations forces $S' = \frac{1}{2}(J + J^T)$ and $A' = \frac{1}{2}(J - J^T)$. So the decomposition is not a choice.

Apply it to the Jacobian of a velocity field. An antisymmetric matrix has zeros on its diagonal ($A_{ii} = -A_{ii}$ forces $A_{ii} = 0$), so $\text{tr} A = 0$ and

$$\nabla \cdot \mathbf{F} = \text{tr} J = \text{tr} S. \quad (0.7.35)$$

All of the divergence lives in the symmetric part. And the antisymmetric part, written out with $J^i_j = \partial_j F^i$, is

$$A = \frac{1}{2} \begin{pmatrix} 0 & \partial_y F^x - \partial_x F^y & \partial_z F^x - \partial_x F^z \\ \partial_x F^y - \partial_y F^x & 0 & \partial_z F^y - \partial_y F^z \\ \partial_x F^z - \partial_z F^x & \partial_y F^z - \partial_z F^y & 0 \end{pmatrix}. \quad (0.7.36)$$

Every entry is one of the three components of (0.7.33), up to a sign. Writing $\omega \equiv \nabla \times \mathbf{F}$, the matrix is exactly

$$A = \frac{1}{2} \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}. \quad (0.7.37)$$

Check one entry to be sure: $A_{21} = \frac{1}{2}(\partial_x F^y - \partial_y F^x) = \frac{1}{2}\omega_z$ by (0.7.32), and that is what sits in row 2, column 1 of (0.7.37). The other five follow the same way.

A matrix of that shape does something recognisable. Apply it to a displacement $\mathbf{h}$:

$$A\mathbf{h} = \frac{1}{2} \begin{pmatrix} \omega_y h^3 - \omega_z h^2 \\ \omega_z h^1 - \omega_x h^3 \\ \omega_x h^2 - \omega_y h^1 \end{pmatrix} = \frac{1}{2} \omega \times \mathbf{h}. \quad (0.7.38)$$

And $\mathbf{h} \mapsto \boldsymbol{\Omega} \times \mathbf{h}$ is precisely the velocity field of a rigid rotation with angular velocity $\boldsymbol{\Omega}$. Put the pieces together with Chapter 0.6's linearization $\mathbf{F}(\mathbf{p} + \mathbf{h}) = \mathbf{F}(\mathbf{p}) + J\mathbf{h} + o(|\mathbf{h}|)$:

WHAT A FLOW DOES TO A SMALL BLOB

$$\mathbf{F}(\mathbf{p} + \mathbf{h}) = \underbrace{\mathbf{F}(\mathbf{p})}_{\text{translate}} + \underbrace{S\mathbf{h}}_{\text{strain}} + \underbrace{\frac{1}{2}\omega \times \mathbf{h}}_{\text{rotate}} + o(|\mathbf{h}|)$$

A small ball of tracer particles is **carried** along at $\mathbf{F}(\mathbf{p})$; **stretched or squeezed** along the eigenvectors of the symmetric part S (which is symmetric, so Chapter 0.5's spectral theorem gives it an orthonormal eigenbasis, and the eigenvalues are the stretching rates along those axes — their sum being the divergence, by (0.7.35)); and **rotated rigidly** at angular velocity $\frac{1}{2}\nabla \times \mathbf{F}$.

So: **the curl is twice the local angular velocity of the flow**. The factor of two is not a convention anybody chose — it fell out of (0.7.38).

This is the cleanest possible statement of what divergence and curl are. They are the two invariant pieces of the Jacobian: the trace, and the antisymmetric part. Note also what they are *not*: they are not everything. The traceless symmetric part of S is left over, it describes pure shear, and it is invisible to both operators. A field can therefore deform a blob dramatically while having zero divergence and zero curl. The figure below is built to show you exactly that, because no static picture can.

4.4 · Why the curl is a vector only in three dimensions

Count. An antisymmetric $n \times n$ matrix is determined by its entries strictly above the diagonal, of which there are $\frac{1}{2}n(n - 1)$. A vector in n dimensions has n components. Those two counts agree when

$$\frac{1}{2}n(n - 1) = n \quad \iff \quad n - 1 = 2 \quad \iff \quad n = 3. \tag{0.7.39}$$

Only in three dimensions. The table is short and worth memorising:

n	ANTISYMMETRIC COMPONENTS	VECTOR COMPONENTS	CURL IS...
2	1	2	a <i>scalar</i>
3	3	3	a <i>vector</i> — the lucky coincidence
4	6	4	an antisymmetric <i>matrix</i> , irreducibly

In two dimensions there is one number and it does not point anywhere, which is why §2 wrote $(\nabla \times \mathbf{F})_z$ and treated it as a scalar. In four dimensions there are six components and no vector to pack them into, so the antisymmetric object has to stay a matrix.

FORESHADOWING THAT IS EXACT, NOT ANALOGICAL

Spacetime is four-dimensional. So when Chapter 2.6 antisymmetrises the derivative of the gauge potential A_μ , the result cannot be a vector; it must be the antisymmetric tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

with $\frac{1}{2} \cdot 4 \cdot 3 = 6$ independent components. And the electromagnetic field has exactly six numbers at each point: three of $\mathbf{E}$ and three of $\mathbf{B}$. They are not two vectors. They are the six entries of one antisymmetric matrix, split into a "time-space" block and a "space-space" block by an observer's choice of what time means — which is why a moving observer sees electric and magnetic fields mix into each other.

The reason $\mathbf{B}$ ever looked like a vector is (0.7.39): in the three-dimensional space-space block, and only there, an antisymmetric matrix has as many entries as a vector, so you can repackage it. The magnetic field is not a vector; it is a 3×3 antisymmetric matrix that has been allowed to impersonate one. Chapter 2.6 stops the impersonation, and the cross product goes with it — $\mathbf{a} \times \mathbf{b}$ is a three-dimensional accident for exactly the same reason.

▼ Grind box — the index form, ϵ_{ijk} , and the dimension count again

Define the **Levi-Civita symbol** ϵ_{ijk} to be $+1$ if (ijk) is a cyclic permutation of (123) , -1 if it is an odd permutation, and 0 if any index repeats. Then (0.7.33) is the single line

$$(\nabla \times \mathbf{F})_i = \epsilon_{ijk} \partial_j F^k,$$

summed over j and k . Check the case $i = 3$: the nonzero terms are $\epsilon_{312} = +1$ and $\epsilon_{321} = -1$, giving $\partial_1 F^2 - \partial_2 F^1$, which is (0.7.32). ✓

The contraction identity. We will use

$$\epsilon_{ijk} \epsilon_{klm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl},$$

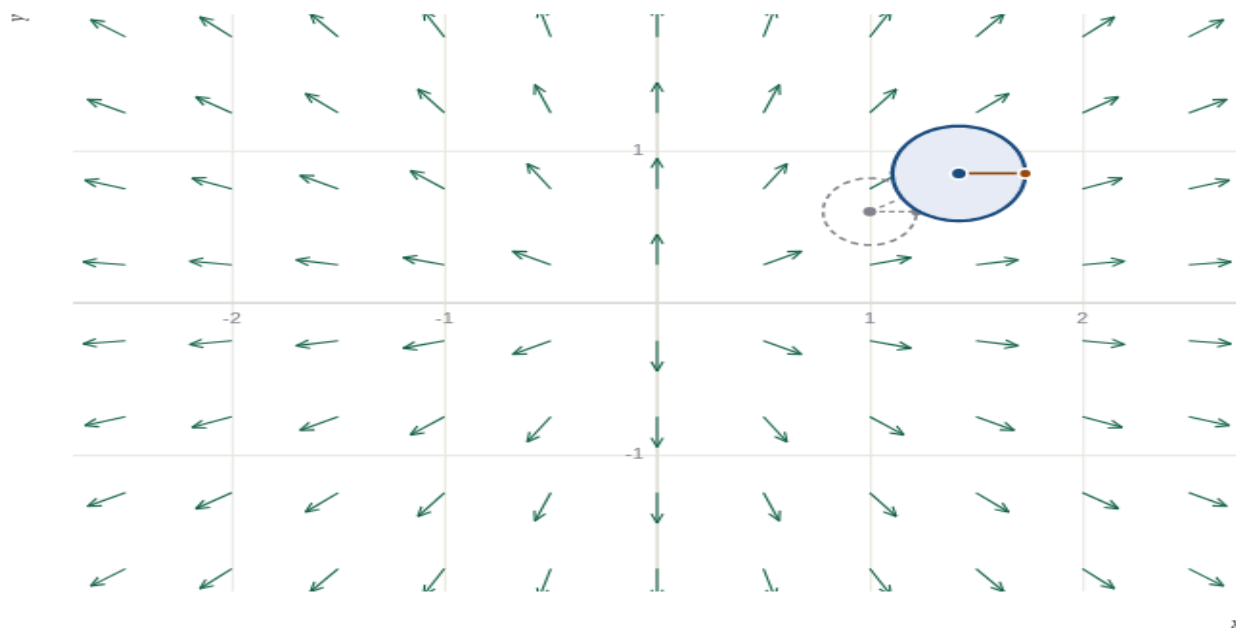
and here is the argument rather than an assertion. Both sides are antisymmetric under $i \leftrightarrow j$ and under $l \leftrightarrow m$, and both vanish unless $\{i, j\} = \{l, m\}$ as sets with $i \neq j$ (on the left, because the sum over k needs $k \notin \{i, j\}$ and $k \notin \{l, m\}$, and in three dimensions that leaves at most one k). So it is enough to check one representative case, $i = l = 1, j = m = 2$: the left side is $\epsilon_{12k} \epsilon_{k12} = \epsilon_{123} \epsilon_{312} = 1$, and the right side is $\delta_{11} \delta_{22} - \delta_{12} \delta_{21} = 1$. ✓ Antisymmetry propagates the check to every other case.

The antisymmetric part in index form. With $\omega = \nabla \times \mathbf{F}$,

$$\epsilon_{ijk} \omega_k = \epsilon_{ijk} \epsilon_{klm} \partial_l F^m = \partial_i F^j - \partial_j F^i = -2A_{ij},$$

using $A_{ij} = \frac{1}{2}(\partial_j F^i - \partial_i F^j)$ from (0.7.34). So $A_{ij} = -\frac{1}{2}\epsilon_{ijk} \omega_k$, which is (0.7.37) written compactly, and $\omega_k = -\epsilon_{kij} A_{ij}$ inverts it.

And now the dimension count, structurally. The repackaging works because ϵ_{ijk} has three indices and can convert a two-index antisymmetric object into a one-index object. In n dimensions the Levi-Civita symbol has n indices, so it converts an antisymmetric 2-index object into an antisymmetric $(n - 2)$ -index one. For $n = 3$ that is a vector; for $n = 4$ it is another antisymmetric matrix, which is why Chapter 2.6's $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ (the "dual" field, which swaps $\mathbf{E}$ and $\mathbf{B}$) is a matrix and not a vector. The cross product, the curl-as-vector, and the fact that a rotation in three dimensions has an *axis* are all the same coincidence: $\frac{1}{2}n(n - 1) = n$ at $n = 3$. In four dimensions a rotation has no axis — it has an invariant plane — and Chapter 2.2's Lorentz transformations are exactly such rotations.



source (x, y)

rotation (-y, x)

shear (y, x)

mixed (xy, x+y)

point x

point y

elapsed t t = 0.35 $\nabla \cdot \mathbf{F} = +2.000$ · measured $\ln(A/A_0)/t = +2.000$

$(\nabla \times \mathbf{F})_z = +0.000$ · measured $2\vartheta/t = -0.000$

blob: area $\times 2.014$, stretched $\times 1.42$ / $\times 1.42$, turned -0.0°

What the flow does to a blob. Green arrows are the field, all drawn the same length so that only their *direction* is being claimed. The dashed grey circle is a ring of tracer particles placed around the sample point; the solid blue shape is where the flow has carried them after a time t (integrated numerically, not linearised — so this is the real flow, and the linear theory of §4.3 is what is being tested). The short radial spokes mark the same material particle before and after, so you can see rotation directly. **The two readouts are measurements, not restatements.** The blob's area ratio gives $\ln(A/A_0)/t$, which by (0.7.27) must equal the divergence; the best-fit rotation angle ϑ of the deformed blob gives $2\vartheta/t$, which by (0.7.38) must equal the curl. Compare each with the analytic value printed beside it. **Now work through the presets.** *Source*: the circle grows and stays a circle — pure divergence, 2, and no rotation. *Rotation*: it turns without changing size or shape — pure curl, 2, zero divergence. *Shear*: **both readouts are zero, and the circle is visibly crushed into an ellipse.** That is the point of the figure. Divergence and curl are independent, and between them they do not capture everything a flow does — the leftover is the traceless symmetric part, and the ellipse's axes are its eigenvectors, here $(1, 1)$ and $(1, -1)$. Note that this field is $\nabla(xy)$, so it is conservative *and* incompressible, and it still shreds the blob. *Mixed*: here $\nabla \cdot \mathbf{F} = y + 1$ and $(\nabla \times \mathbf{F})_z = 1 - x$, so you can dial each independently by dragging — try $(1, -1)$ for zero-and-zero, $(1, 1)$ for pure divergence, $(-1, -1)$ for pure curl. For this field the measured and analytic numbers differ, because the blob travels to places with different divergence during the time t ; drag t toward zero and watch them converge. That gap is the limit in (0.7.17), made visible.

5 · The three theorems are one theorem

5.1 · The statements

Here are the three results, so you know where we are going.

The divergence theorem (Gauss, Ostrogradsky). For a region V with closed boundary surface ∂V , outward normal:

$$\int_V (\nabla \cdot \mathbf{F}) \, dV = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}. \quad (0.7.40)$$

Stokes' theorem. For a surface S with boundary curve ∂S , oriented by the right-hand rule relative to the surface normal:

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}. \quad (0.7.41)$$

Green's theorem. For a region R in the plane with boundary curve ∂R traversed counterclockwise, and any two functions P, Q :

$$\int_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy = \oint_{\partial R} (P \, dx + Q \, dy). \quad (0.7.42)$$

Green's theorem is not independent: put $\mathbf{F} = (P, Q, 0)$ and take S to be the flat region R with $\hat{\mathbf{n}} = \hat{\mathbf{z}}$. Then (0.7.32) makes the left side of (0.7.41) into the left side of (0.7.42), and $\mathbf{F} \cdot d\mathbf{r} = P \, dx + Q \, dy$ makes the right sides agree.

Green's theorem is the special case of Stokes' theorem for a flat surface. Two to go.

5.2 · The divergence theorem, derived by cancellation

Chop the region V into N small cells $V_1, \dots, V_N$ — think of a fine grid of boxes. For each cell, the definition of divergence (0.7.17) says that its outward flux is the divergence at an interior point times its volume, up to something smaller:

$$\oint_{\partial V_k} \mathbf{F} \cdot d\mathbf{A} = (\nabla \cdot \mathbf{F})(\mathbf{p}_k) |V_k| + o(|V_k|). \quad (0.7.43)$$

Now sum over all cells. Look at the two sides separately.

The right side is a Riemann sum. $\sum_k (\nabla \cdot \mathbf{F})(\mathbf{p}_k) |V_k|$ is exactly the tagged sum that Chapter 0.2 defined the integral to be the limit of, so as the cells shrink it converges to $\int_V (\nabla \cdot \mathbf{F}) \, dV$.

The left side collapses. Every face of every cell is either an interior face, shared with a neighbouring cell, or a piece of the outer boundary ∂V . Consider an interior face shared by cells k and ℓ . The *same* field $\mathbf{F}$ is integrated over the *same* patch of surface twice — once with the outward normal of cell k , once with the outward normal of cell ℓ . Those two

normals are opposite, because outward from k is inward to ℓ . So the two contributions are $+\int \mathbf{F} \cdot \hat{\mathbf{n}} \, dA$ and $-\int \mathbf{F} \cdot \hat{\mathbf{n}} \, dA$, and they cancel *exactly*. Not approximately: they are the same integral with opposite signs.

Every interior face is shared by exactly two cells, so every interior contribution cancels, and what survives is the sum over faces lying on ∂V — which is precisely $\oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}$. Equating the two limits:

$$\boxed{\int_V (\nabla \cdot \mathbf{F}) \, dV = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}.} \quad (0.7.44)$$

YOU HAVE SEEN THIS ARGUMENT BEFORE

Chapter 0.2 proved the Fundamental Theorem like this: partition $[a, b]$, telescope $G(b) - G(a) = \sum_i [G(x_i) - G(x_{i-1})]$, notice that every interior partition point appears twice with opposite signs and cancels, and conclude that *only the boundary survives*.

That is the paragraph you just read, with "partition point" replaced by "face". In one dimension the cells are intervals and their boundaries are points; in three dimensions the cells are boxes and their boundaries are squares. The mechanism — **add up local changes, watch the interior cancel in pairs, keep what is left on the boundary** — is identical, and it is the only mechanism in this chapter.

5.3 • Stokes' theorem, by the identical move

Tile the surface S with small patches $S_1, \dots, S_N$, each with its own boundary loop oriented consistently with the surface normal. By the definition of curl (0.7.29), each patch satisfies

$$\oint_{\partial S_k} \mathbf{F} \cdot d\mathbf{r} = (\nabla \times \mathbf{F})(\mathbf{p}_k) \cdot \hat{\mathbf{n}}_k |S_k| + o(|S_k|). \quad (0.7.45)$$

Sum. The right side is a Riemann sum for $\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{A}$. On the left, every interior edge is shared by exactly two patches — and because the two patches are traversed with a consistent orientation, they walk that shared edge in *opposite directions*. By the orientation rule of §2.1's grind box, the two line integrals over that edge are negatives of each other and cancel exactly. Only edges on the outer boundary ∂S survive, giving (0.7.41). ■

The consistency of the orientations is where the right-hand rule earns its keep: it is the rule that guarantees adjacent patches disagree about the direction of their shared edge. If you have ever wondered why Stokes' theorem comes with a hand attached, that is why.

5.4 • One theorem

Line up everything this book has proved that has this shape.

THEOREM	REGION M	BOUNDARY ∂M	THE THING ω	ITS DERIVATIVE $d\omega$
FTC (0.2)	interval	two endpoints	f	$f' dx$
Gradient (§2.2)	curve	two endpoints	ϕ	$\nabla\phi \cdot d\mathbf{r}$
Stokes / Green	surface	closed curve	$\mathbf{F} \cdot d\mathbf{r}$	$(\nabla \times \mathbf{F}) \cdot d\mathbf{A}$
Divergence	volume	closed surface	$\mathbf{F} \cdot d\mathbf{A}$	$(\nabla \cdot \mathbf{F}) dV$

Every row says the same thing, and the operators ∇ , $\nabla \times$, $\nabla \cdot$ are three appearances of one operation applied to objects of three different types. Chapter 3.5 will build the objects properly — they are called **differential forms**, the operation is the **exterior derivative** d , and ω ranges over forms of degree 0, 1, 2 — and then all four rows become one equation:

$$\boxed{\int_M d\omega = \oint_{\partial M} \omega.} \quad (0.7.46)$$

We are quoting the general form, not deriving it: the machinery needed to define ω and $d\omega$ on a manifold of arbitrary dimension is Chapter 3.5's job, and doing it here would be building a cathedral to hang a door. But nothing in the table above is being taken on faith — all four rows have been derived, in Chapter 0.2 and in this chapter, by the same cancellation argument. What Chapter 3.5 adds is notation good enough to say it once.

One structural remark to file. A boundary has no boundary: the boundary of a disc is a circle, and a circle has no endpoints; the boundary of a ball is a sphere, and a sphere has no edge. Symbolically $\partial(\partial M) = \emptyset$. Under the correspondence of the table, that fact about regions must correspond to a fact about derivatives — and it does. It is $d^2 = 0$, and §7 proves the two instances of it you already know.

▼ Grind box — what the cancellation argument does and does not establish

The subdivision argument is the right picture, it is how every working physicist thinks about these theorems, and it is not a proof. Three gaps, in increasing order of seriousness.

(i) Uniformity of the remainder. We wrote $o(|V_k|)$ for each cell and then summed N of them, with $N \rightarrow \infty$. A sum of many small errors need not be small. What is needed is that the error is $\varepsilon |V_k|$ with a *single* ε that works for all cells at once and tends to zero as the grid refines; then the total error is $\varepsilon |V| \rightarrow 0$. Uniformity of that kind is supplied by continuity of the partial derivatives on a closed bounded region — the same hypothesis, and the same reason, as in Chapter 0.6 §8's grind box.

(ii) The boundary is a staircase. Cells that straddle ∂V are not fully inside, and a grid of boxes approximates a curved boundary by a staircase whose *area* does not in general converge to the true area even when the enclosed volume does. Repairing this honestly means either using cells adapted to the boundary or a limiting argument that controls the boundary layer's contribution — true for piecewise-smooth boundaries, and genuinely false for sufficiently ugly ones.

(iii) **We assumed the definitions converge.** (0.7.43) is the definition of divergence read backwards, which presumes the limit in (0.7.17) exists and is approached uniformly over the region. For C^1 fields it does, by the argument in §3.3's grind box; for fields that are merely differentiable it can fail.

The honest statement of the theorem is therefore: for a compact region with piecewise-smooth boundary and a continuously differentiable field, (0.7.44) holds. That covers everything in this book. What you should take from the argument is the *mechanism* — interior faces cancel in pairs — because the mechanism is what generalises to Chapter 3.5's manifolds, where there is no grid of boxes to draw and the cancellation is enforced algebraically by the antisymmetry of forms instead.

6 · The continuity equation

Now the single most reused equation in physics, and it is three lines from §5.

Let $\rho(\mathbf{x}, t)$ be the density of some substance — charge per unit volume, mass per unit volume, probability per unit volume — and let $\mathbf{J}(\mathbf{x}, t)$ be its **current density**, the vector field whose flux through a surface is the rate at which the substance crosses it (§3.1). Fix a region V , *fixed in space and not moving*. The amount inside is

$$Q_V(t) = \int_V \rho(\mathbf{x}, t) \, dV. \quad (0.7.47)$$

Now impose the physical assumption, which is the entire content of the derivation: **the substance is neither created nor destroyed**, so the only way the amount inside V can change is by flowing through the boundary. Outflow decreases it, hence the minus sign:

$$\frac{dQ_V}{dt} = - \oint_{\partial V} \mathbf{J} \cdot d\mathbf{A}. \quad (0.7.48)$$

Handle each side. On the left, V is fixed, so the time derivative passes through the integral and lands on the integrand as a partial derivative (differentiating under the integral sign, legitimate here for the same reason as in Chapter 0.2 §6: the integrand and its t -derivative are continuous and the domain is fixed). On the right, apply the divergence theorem (0.7.44):

$$\int_V \frac{\partial \rho}{\partial t} \, dV = - \int_V \nabla \cdot \mathbf{J} \, dV \quad \implies \quad \int_V \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} \right) \, dV = 0. \quad (0.7.49)$$

And now the step that does the real work: V **was arbitrary**. We never said which region. So the integral of that bracket vanishes over every region whatsoever, and a continuous function with that property is zero.

LEMMA — VANISHING INTEGRALS OVER ALL REGIONS

If g is continuous and $\int_V g \, dV = 0$ for every region V , then $g \equiv 0$.

Proof. Suppose $g(\mathbf{p}) > 0$ for some $\mathbf{p}$. By continuity there is a ball B around $\mathbf{p}$ on which $g > \frac{1}{2}g(\mathbf{p})$, and then $\int_B g \, dV \geq \frac{1}{2}g(\mathbf{p}) |B| > 0$, contradicting the hypothesis for $V = B$. The same argument with signs reversed handles $g(\mathbf{p}) < 0$. Hence g vanishes everywhere. ■

Continuity is essential and is doing exactly the work you would expect: it is what stops the function from being nonzero on a set too small for any ball to notice.

Applying the lemma to (0.7.49):

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.} \quad (0.7.50)$$

6.1 · What it says, and why "local" is the important word

Read it at a point. The density here goes down exactly as fast as the current here diverges — as fast as stuff flows away from here. That is a much stronger statement than "the total amount in the universe is constant". Global conservation would permit a charge to vanish in London and simultaneously appear in Sydney; the books would balance. (0.7.50) forbids it: to leave a region the substance must cross the boundary, and to get anywhere it must travel through the intervening space.

WHAT A CONSERVATION LAW ACTUALLY IS

A local conservation law is a continuity equation. " X is conserved" means there exists a current $\mathbf{J}_X$ such that $\frac{\partial \rho_X}{\partial t} + \nabla \cdot \mathbf{J}_X = 0$. The global statement is the integrated version, not the definition.

The global version follows by integrating over all space. Take V to be a ball of radius R and integrate (0.7.50); the divergence theorem turns the second term back into a surface flux:

$$\frac{d}{dt} \int_{|\mathbf{x}| < R} \rho \, dV = - \oint_{|\mathbf{x}|=R} \mathbf{J} \cdot d\mathbf{A}. \quad (0.7.51)$$

The surface area grows like R^2 , so if $\mathbf{J}$ falls off faster than $1/R^2$ the right side tends to zero as $R \rightarrow \infty$, and the total amount in the universe is constant. Global conservation is a *consequence* of local conservation plus a boundary condition at infinity — and when the boundary condition fails, as it does for a radiating system, it fails for good physical reasons.

6.2 · Where it turns up

Constantly. A partial list, all of which this book reaches:

- **Charge, in electromagnetism (Chapter 2.6).** ρ is charge density and $\mathbf{J}$ the electric current. Problem 4 shows that Maxwell's equations do not merely permit this equation — they *force* it, and the mechanism is §7's identity $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.
- **Probability, in quantum mechanics (Chapter 4.4).** With $\rho = |\psi|^2$ and a current built from ψ and $\nabla\psi$, the Schrödinger equation implies exactly (0.7.50). That is what "the wavefunction stays normalised" means locally: probability does not teleport.
- **Noether currents (Chapters 1.4 and 5.2).** Noether's theorem does not produce a conserved number; it produces a conserved *current* satisfying $\partial_\mu J^\mu = 0$, of which (0.7.50) is the component-by-component reading. Every continuous symmetry gives one.
- **Energy and momentum, in general relativity (Chapter 3.6).** $\nabla_\mu T^{\mu\nu} = 0$ is this equation for the energy-momentum tensor, with an index left over because energy and momentum travel together. In general relativity it is not imposed — it is forced by a geometric identity, which is $d^2 = 0$ yet again.
- **Diffusion (the callout at the end of §7).** Combine it with Fick's law and out falls the diffusion equation.

One notational preview, because it explains why the equation looks lopsided. In relativity ρ and $\mathbf{J}$ are not separate objects: they are the four components of a single four-vector $J^\mu = (\rho c, \mathbf{J})$, and (0.7.50) is

$$\partial_\mu J^\mu = 0, \quad (0.7.52)$$

a four-dimensional divergence set to zero. The time derivative and the spatial divergence are the same operation, split up by an observer's choice of what counts as time. Chapter 2.4 makes this precise; note in passing that it is the same "one object, split by an observer" story as $\mathbf{E}$ and $\mathbf{B}$ in §4.4.

7 · Second-derivative identities, and the Laplacian

Two identities, both one line of Clairaut, both structurally important out of all proportion to their difficulty.

7.1 · The curl of a gradient vanishes

Take $\mathbf{F} = \nabla\phi$, so $F^i = \partial_i\phi$, and put it into (0.7.33). The z -component is

$$(\nabla \times \nabla\phi)_z = \partial_x(\partial_y\phi) - \partial_y(\partial_x\phi) = 0 \quad (0.7.53)$$

by Clairaut's theorem (Chapter 0.6 §6.1), which says mixed partials commute when they are continuous. The other two components are the same statement with the labels rotated. Hence

$$\boxed{\nabla \times \nabla\phi = \mathbf{0}} \quad \text{for every twice-differentiable } \phi. \quad (0.7.54)$$

That completes the chain (A) $\Rightarrow$ (D) of §2.3: a gradient field is curl-free, always, everywhere, with no hypothesis on the domain. It is the converse that needed the domain to be simply connected, and §2.4 is why.

7.2 · The divergence of a curl vanishes

Write it out. With $\mathbf{F} = \nabla \times \mathbf{A}$, (0.7.21) gives

$$\begin{aligned}\nabla \cdot (\nabla \times \mathbf{A}) &= \partial_x (\partial_y A^z - \partial_z A^y) \\ &\quad + \partial_y (\partial_z A^x - \partial_x A^z) \\ &\quad + \partial_z (\partial_x A^y - \partial_y A^x).\end{aligned}\tag{0.7.55}$$

Six terms. Pair them off: $\partial_x \partial_y A^z$ against $-\partial_y \partial_x A^z$, $-\partial_x \partial_z A^y$ against $+\partial_z \partial_x A^y$, and $\partial_y \partial_z A^x$ against $-\partial_z \partial_y A^x$. Each pair cancels by Clairaut. So

$$\boxed{\nabla \cdot (\nabla \times \mathbf{A}) = 0} \quad \text{for every twice-differentiable } \mathbf{A}.\tag{0.7.56}$$

7.3 · Why the vector potential exists

Maxwell's equations contain $\nabla \cdot \mathbf{B} = 0$: there are no magnetic charges. Compare that with (0.7.56). Any field written as a curl automatically satisfies it — so the ansatz

$$\mathbf{B} = \nabla \times \mathbf{A}\tag{0.7.57}$$

is *consistent* with Maxwell for any $\mathbf{A}$ whatsoever. The converse — that every divergence-free field can be written this way — is true on regions without holes, and is the **Poincaré lemma**, which we quote here and Chapter 3.5 proves. (The parallel with §2 is exact and not accidental: there, curl-free implied "is a gradient" only on a simply connected domain; here, divergence-free implies "is a curl" only on a domain without cavities. Both are instances of one statement about the topology of the region, which Chapter 3.5 calls de Rham cohomology and which is measuring, in each case, the difference between the fields that are *closed* and those that are *exact*.)

Now put the two identities side by side, because together they are the skeleton of gauge theory:

- (0.7.56) is why the vector potential *exists*: it makes $\mathbf{B} = \nabla \times \mathbf{A}$ compatible with $\nabla \cdot \mathbf{B} = 0$.
- (0.7.54) is why the vector potential is *not unique*: replacing $\mathbf{A} \rightarrow \mathbf{A} + \nabla \chi$ for any function χ leaves $\mathbf{B} = \nabla \times \mathbf{A}$ unchanged, since $\nabla \times \nabla \chi = \mathbf{0}$.

That second bullet is **gauge freedom**, and it is the seed of Chapter 6.3, where demanding that χ be allowed to vary from point to point *generates* the electromagnetic interaction. Two one-line consequences of Clairaut, and between them they set up the entire structure of the Standard Model.

7.4 · Both identities are $d^2 = 0$

Line up the operators by what they act on:

$$\text{functions} \xrightarrow{\nabla} \text{vector fields} \xrightarrow{\nabla \times} \text{vector fields} \xrightarrow{\nabla \cdot} \text{functions}.\tag{0.7.58}$$

Composing two consecutive arrows gives zero, both times: $\nabla \times \nabla = 0$ and $\nabla \cdot \nabla \times = 0$. In Chapter 3.5's language the three objects are differential forms of degree 0, 1, 2, all three arrows are the same operator d , and the two identities are the single statement

$$d \circ d = 0. \quad (0.7.59)$$

Which is, by the correspondence of §5.4, the algebraic shadow of $\partial(\partial M) = \emptyset$: a boundary has no boundary. We are flagging that as a promise, not a derivation — Chapter 3.5 makes it precise. But you can already see that the two facts you just proved are not two facts.

7.5 · The Laplacian

The remaining composition of two first-order operators is the one that is *not* zero: divergence of a gradient. Define the **Laplacian**

$$\nabla^2 \phi \equiv \nabla \cdot \nabla \phi = \partial_i \partial_i \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}. \quad (0.7.60)$$

Structurally it is the *trace of the Hessian* of Chapter 0.6 §6 — and since the trace is basis-independent (Chapter 0.4 §6), so is the Laplacian, even though the formula above mentions a particular set of axes.

One reading is immediate from §3: $\nabla^2 \phi$ is the flux of $\nabla \phi$ per unit volume, so it is positive where the gradient field converges on the point — where the surroundings are pushing inward. The sharper reading is the following, and it is the one to carry.

The mean-value reading

Expand ϕ around $\mathbf{p}$ to second order, using Chapter 0.6 §6.2's multivariable Taylor:

$$\phi(\mathbf{p} + \mathbf{h}) = \phi(\mathbf{p}) + h^i \partial_i \phi + \frac{1}{2} h^i h^j \partial_i \partial_j \phi + O(|\mathbf{h}|^3). \quad (0.7.61)$$

Average this over the sphere $|\mathbf{h}| = a$. Three averages are needed and all three are pure symmetry.

First order. $\langle h^i \rangle = 0$, because $\mathbf{h}$ and $-\mathbf{h}$ both lie on the sphere and cancel.

Second order. $\langle h^i h^j \rangle$ is a symmetric array built from a sphere, and a sphere has no preferred direction, so the only array it can be is a multiple of δ^{ij} : $\langle h^i h^j \rangle = c \delta^{ij}$. Fix c by taking the trace of both sides — set $j = i$ and sum: the left becomes $\langle |\mathbf{h}|^2 \rangle = a^2$, the right becomes $3c$ in three dimensions. Hence $c = a^2/3$ and

$$\langle h^i h^j \rangle = \frac{a^2}{3} \delta^{ij}. \quad (0.7.62)$$

Third order. Cubic in $\mathbf{h}$, so it changes sign under $\mathbf{h} \rightarrow -\mathbf{h}$ and averages to zero. The first surviving correction is therefore quartic.

Putting them together, and noting $\delta^{ij}\partial_i\partial_j\phi = \nabla^2\phi$:

$$\bar{\phi}_{\text{sphere}}(a) - \phi(\mathbf{p}) = \frac{a^2}{6} \nabla^2\phi(\mathbf{p}) + O(a^4). \quad (0.7.63)$$

WHAT THE LAPLACIAN MEASURES

$\nabla^2\phi$ at a point tells you how much the field there differs from its average on a small sphere around it — positive if the point is a dip relative to its surroundings, negative if it is a bump. In n dimensions the coefficient is $a^2/2n$; in the plane, $a^2/4$.

An immediate consequence, free of charge. If $\nabla^2\phi = 0$ everywhere — such a ϕ is called **harmonic** — then the value at every point equals its average over every small sphere around it. So ϕ can have no strict interior maximum: at a strict maximum, every nearby value is smaller, so the spherical average would be strictly *less* than the central value, contradicting equality. Hence a harmonic function attains its extremes on the boundary. That is the **maximum principle**, it is why you cannot trap a charged particle in a static electric field (Earnshaw's theorem — the potential energy is harmonic in charge-free space, so it has no minimum to sit in), and it fell out of a symmetry argument about $\langle h^i h^j \rangle$.

Poisson's equation

Combine §3 with §2. Gauss's law in differential form says $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ (we take this as the physical input; Chapter 2.6 derives it). And the electric field is minus the gradient of a potential, $\mathbf{E} = -\nabla\phi$, which is possible because the electrostatic field is curl-free and §2.3 then supplies ϕ . Substituting one into the other:

$$\nabla^2\phi = -\frac{\rho}{\epsilon_0}. \quad (0.7.64)$$

Poisson's equation, and with $\rho = 0$ it is **Laplace's equation** $\nabla^2\phi = 0$. Read it through (0.7.63): charge density is exactly the amount by which the potential fails to equal its own local average. Where there is no charge, the potential is the smoothest interpolation of its boundary values that exists.

The same equation governs Newtonian gravity, $\nabla^2\Phi = 4\pi G\rho$, and that is not a coincidence of two inverse-square forces — Chapter 3.6 will show that the Einstein field equations reduce to precisely this in the weak-field, slow-motion limit, with Φ the 00-component of the metric perturbation. Poisson's equation is what general relativity looks like when you squint.

▼ **Grind box** — $\nabla^2(1/r) = 0$, and the 4π that has gone missing

For a function of r alone, the spherical divergence formula from §3's grind box gives

$$\nabla^2 f(r) = \nabla \cdot (f'(r)\hat{\mathbf{r}}) = \frac{1}{r^2} \frac{d}{dr} (r^2 f'(r)).$$

Take $f = 1/r$, so $f' = -1/r^2$ and $r^2 f' = -1$, a constant:

$$\nabla^2 \frac{1}{r} = \frac{1}{r^2} \frac{d}{dr}(-1) = 0 \quad (r \neq 0).$$

So the Coulomb potential is harmonic everywhere except at the source — consistent with (0.7.64), since $\rho = 0$ away from the point charge.

Now the trouble. Apply the divergence theorem to $\mathbf{E} = -\nabla(1/r) = \hat{\mathbf{r}}/r^2$ on a ball of radius R centred at the origin. The left side is $\int_V \nabla \cdot \mathbf{E} \, dV = 0$, since the integrand vanishes at every point where it is defined. The right side is

$$\oint_{|\mathbf{r}|=R} \frac{\hat{\mathbf{r}}}{r^2} \cdot \hat{\mathbf{r}} \, dA = \frac{1}{R^2} \cdot 4\pi R^2 = 4\pi,$$

independent of R . Zero on one side, 4π on the other. Something is wrong, and it is not the theorem: the theorem requires $\mathbf{E}$ to be continuously differentiable *everywhere inside* V , and this one is not even defined at the origin. The hypothesis fails, so the conclusion is not owed to us.

The repair is to stop pretending the origin is not there and write

$$\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} = 4\pi \delta^3(\mathbf{r}),$$

where δ^3 is an object that is zero everywhere except the origin and integrates to one. No function does that, which is why δ is not a function but a **distribution**, and why Chapter 0.9 exists. The 4π is not a fudge factor: it is the surface area of the unit sphere, and it is there because the flux computation above returned 4π for every R . Notice the pattern with §2.4, where a deleted point produced a circulation of 2π — the circumference of the unit circle. In both cases the removed point is doing all the work, and the number the integral returns is counting it.

FAMILIAR GROUND

You have been using this chapter's two central objects for your whole career, under other names.

Fick's first law says that a solute moves down its concentration gradient:

$$\mathbf{J} = -D \nabla C.$$

Every symbol is now something you have derived. C is a scalar field. ∇C points in the direction of steepest increase of concentration and has magnitude equal to the steepness (Chapter 0.6 §3.1), so $-\nabla C$ points downhill — the minus sign is the whole physical claim. $\mathbf{J}$ is a current density in the sense of §3.1: its flux through a membrane is the number of molecules crossing per second. And D , the diffusion coefficient, must have units of $\text{length}^2/\text{time}$ for the equation to balance, which is the single most useful thing about it.

Fick's second law is not a second law. It is the first law fed into §6. Diffusing solute is neither created nor destroyed, so the continuity equation (0.7.50) applies:

$$\frac{\partial C}{\partial t} + \nabla \cdot \mathbf{J} = 0 \quad \implies \quad \frac{\partial C}{\partial t} = -\nabla \cdot (-D \nabla C).$$

With D constant it comes out of the divergence, and $\nabla \cdot \nabla$ is the Laplacian:

$$\boxed{\frac{\partial C}{\partial t} = D \nabla^2 C.}$$

Two lines. That is the **diffusion equation**, identical to the heat equation, and §7.5 tells you what it means without solving it: *the concentration at a point rises exactly in proportion to the amount by which it falls short of the average concentration on a small sphere around it*. Diffusion is a field chasing its own neighbourhood average. Every qualitative fact about it — that peaks flatten, that dips fill in, that sharp edges blur fastest because that is where $\nabla^2 C$ is largest — reads straight off (0.7.63).

How far, how fast. The structure of the equation fixes the scaling with no solution required. By dimensional analysis (Chapter 0.3 §5), the only length that can be assembled from D and an elapsed time t is

$$\ell \sim \sqrt{Dt} \quad \iff \quad t \sim \frac{\ell^2}{D}.$$

Diffusion time grows as the *square* of the distance, and that quadratic is the reason tissue is organised the way it is. Take $D \approx 10^{-6} \text{ cm}^2/\text{s}$, a reasonable figure for a small drug in tissue (a small molecule in free water is nearer 10^{-5} ; a monoclonal antibody in tumour stroma is nearer 10^{-7}):

DISTANCE	$t = \ell^2/D$
100 μm	100 s — under two minutes
1 mm	10^4 s — about 3 hours
1 cm	10^6 s — about 12 days

Ten times further, a hundred times longer. Diffusion is superb over cellular distances and catastrophic over organ distances, which is why every organism above a millimetre or so has a circulatory system: bulk flow to cross the centimetres, diffusion for the last hundred microns.

The viable rim. Now do the calculation you actually care about. Oxygen diffuses from a capillary into tissue that consumes it at roughly a constant rate k per unit volume. In steady state $\frac{\partial C}{\partial t} = 0$ and the equation becomes $D \nabla^2 C = k$; in one dimension, measuring x from the vessel wall and requiring the flux to vanish at the far edge $x = L$ of the supplied region, $C'' = k/D$ integrates to

$$C(x) = C_0 - \frac{k}{2D} x (2L - x), \quad \text{and } C(L) = 0 \implies L = \sqrt{\frac{2DC_0}{k}}.$$

Read the answer: C_0/k is the time the tissue would take to consume its dissolved oxygen if the supply were cut, so $L = \sqrt{2D\tau}$ with τ that time — the same $\sqrt{Dt}$, wearing a clinical hat. Tissue oxygen stores last of order $\tau \approx 2\text{--}10$ s, and oxygen in tissue has $D \approx 2 \times 10^{-5} \text{ cm}^2/\text{s}$, giving $L \approx 90\text{--}200 \mu\text{m}$. The measured viable rim around a capillary in a tumour cord is 100–150 μm , beyond which the centre is necrotic — the observation Thomlinson and Gray made in 1955, and an experimental fact we quote rather than derive. The order of magnitude is not a coincidence; it is (0.7.50) plus Fick.

The same equation, with a much smaller D and a binding term, is drug penetration into a tumour; it is why an antibody can saturate the perivascular cuff and never reach the core, and why penetration depth scales as the square root of everything you can change.

Now the sting. Write the diffusion equation next to the equation Chapter 4.4 will derive for a free quantum particle:

$$\frac{\partial C}{\partial t} = D \nabla^2 C \quad \text{versus} \quad i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi.$$

Divide the second by $i\hbar$ and it reads $\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \nabla^2 \psi$. **It is the diffusion equation with an imaginary diffusion coefficient, $D \rightarrow i\hbar/2m$.** One equation, one factor of i . Diffusion spreads and damps: a peak flattens and never comes back. Schrödinger spreads and *oscillates*: a wavepacket spreads too, but nothing is lost, and the parts can come back and interfere.

This is the *third* time in Part 0 that an i has converted decay into rotation. Chapter 0.1's callout put $C = C_0 e^{-kt}$ beside $\psi = \psi_0 e^{-iEt/\hbar}$; Chapter 0.3 explained the mechanism, that e^{-kt} shrinks while $e^{i\theta}$ turns at constant length; and now the same substitution has turned an irreversible smearing-out into

unitary quantum evolution. Three times is not a coincidence. Chapter 4.4 says what it is: the i is what makes time evolution a *rotation* in the space of states rather than a contraction, which is exactly what conserving total probability requires. And the substitution $t \rightarrow -i\tau$ that converts one into the other has a name, Wick rotation, and Chapter 5.8 uses it to turn quantum field theory into statistical mechanics — the same trade you already saw at the end of Chapter 0.6.

△ WHY THIS ISN'T OBVIOUS

1. "Curl-free" is a local statement. "Has a potential" is a global one. This is the trap of §2, and it is worth stating as sharply as possible. The field $\mathbf{F} = (-y, x)/(x^2 + y^2)$ is smooth on its whole domain and its curl vanishes at *every single point of that domain*. There is no point at which any local measurement detects anything unusual. And yet no single-valued potential exists, and the circulation around the hole is 2π .

What went wrong is not analysis; it is the *shape of the region*. The theorem "curl-free $\Rightarrow$ conservative" needs every loop to bound a surface inside the domain, and a loop encircling a puncture does not. So the implication holds on a disc and fails on an annulus, with the field unchanged. **A property of the domain, invisible to any local test, has changed the conclusion.**

And physics notices. The Aharonov–Bohm effect is precisely this configuration, made experimental: $\mathbf{B} = \mathbf{0}$ everywhere the electron goes, and the interference pattern moves anyway, because the electron responds to $\oint \mathbf{A} \cdot d\mathbf{r}$ and the solenoid is the hole. This is the first place in the book where a topological fact has a measurable consequence. It is not the last: the quantisation of magnetic charge, the θ -vacuum of QCD (Chapter 6.5), instantons, and the stability of solitons and branes (Chapter 7.7) are all this same observation with more indices.

2. The divergence theorem needs the field to exist everywhere inside. Not almost everywhere. Everywhere. The grind box above worked the example: $\hat{\mathbf{r}}/r^2$ has zero divergence at every point where it is defined, but its flux through any sphere about the origin is 4π . If you apply (0.7.44) without checking the hypothesis you will prove $0 = 4\pi$.

This is not a pathological worry — it is the situation for every point charge and every point mass in physics, which is to say most of the first two years of the subject. The honest resolution is that $\nabla \cdot (\hat{\mathbf{r}}/r^2)$ is not zero but $4\pi\delta^3(\mathbf{r})$, an object that is not a function, and Chapter 0.9 builds it properly. Until then, the rule is: when you use Gauss's theorem, look inside the region and ask what is there.

Notice that both warnings have the same shape. In each case a single removed point carries all the content, and the integral around it returns a number — 2π in the plane, 4π on the sphere — that is measuring the point rather than the field. That is not an analogy. Chapter 3.5's cohomology is the machinery for saying it once.

8 · Worked examples

WORKED EXAMPLE 1 — THE INVERSE-SQUARE LAW IS A FACT ABOUT SPHERES

Given Gauss's law in differential form, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, find the electric field of a spherically symmetric charge distribution of total charge q contained within radius a .

Step 1: symmetry fixes the form of the answer. The charge distribution is unchanged by every rotation about the centre, so the field must be too. A vector field invariant under all such rotations can only point radially — any transverse component would be turned into a different transverse component by some rotation — and its magnitude can only depend on r . Hence $\mathbf{E} = E(r) \hat{\mathbf{r}}$, with one unknown function of one variable instead of three functions of three.

Step 2: apply the divergence theorem to the ball V of radius $r > a$:

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \int_V \nabla \cdot \mathbf{E} dV = \frac{1}{\epsilon_0} \int_V \rho dV = \frac{q}{\epsilon_0}.$$

Step 3: evaluate the flux. On the sphere $\hat{\mathbf{n}} = \hat{\mathbf{r}}$, so $\mathbf{E} \cdot \hat{\mathbf{n}} = E(r)$, which is constant over the surface and comes out of the integral, leaving the area:

$$E(r) \cdot 4\pi r^2 = \frac{q}{\epsilon_0} \quad \implies \quad \boxed{E(r) = \frac{q}{4\pi\epsilon_0 r^2}}.$$

Three lines, and Coulomb's law has been derived rather than postulated. Two things fell out for free. Only the charge *enclosed* appeared, so for $r < a$ the same argument gives $E(r) = q_{\text{enc}}(r)/4\pi\epsilon_0 r^2$ — and a hollow shell, having no enclosed charge, exerts no force anywhere inside it. That is Newton's shell theorem, which cost him a page of classical geometry.

Now the observation that matters. Look at where the r^2 came from. It came from $4\pi r^2$, the *area of a sphere*. Nothing about electricity produced it. The physical input was that flux is conserved — field lines start on charges and go somewhere — and the geometric input was that the same number of lines is spread over a surface whose area grows as r^2 .

So run the argument in n spatial dimensions. The sphere of radius r has area $\Omega_{n-1} r^{n-1}$, where Ω_{n-1} is the area of the unit sphere (a pure number whose value we do not need). Step 3 becomes $E \cdot \Omega_{n-1} r^{n-1} = q/\epsilon_0$, so

$$E(r) \propto \frac{1}{r^{n-1}}, \quad \phi(r) \propto \frac{1}{r^{n-2}} \quad (n > 2), \quad \phi \propto \ln r \quad (n = 2).$$

The inverse-square law is not a law about forces. It is the statement that we live in three spatial dimensions.

That is a testable claim, and Chapter 7.7 tests it. If there are extra spatial dimensions curled up on a scale R , then at separations much smaller than R the flux spreads into *all* the dimensions and gravity falls off as $1/r^{n-1}$ with $n > 3$, while at separations much larger than R the extra directions are used up and the familiar $1/r^2$ returns. Sub-millimetre torsion-balance measurements of the gravitational inverse-square law are

therefore literally searches for extra dimensions, and the bound on R they produce is a bound on the geometry of spacetime — obtained by weighing things very carefully at short range.

WORKED EXAMPLE 2 — TAKING A JACOBIAN APART

For $\mathbf{F} = (x^2y, 3x - z^2, yz)$, compute the divergence and the curl at $\mathbf{p} = (1, 1, 1)$, decompose the Jacobian into its symmetric and antisymmetric parts, and check the identification of §4.3.

The Jacobian. Differentiate each component with respect to each variable, following Chapter 0.6 §2.2 — row i is F^i , column j is ∂_j :

$$J(x, y, z) = \begin{pmatrix} 2xy & x^2 & 0 \\ 3 & 0 & -2z \\ 0 & z & y \end{pmatrix}, \quad J(1, 1, 1) = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}.$$

Divergence is the trace, by (0.7.22):

$$\nabla \cdot \mathbf{F} = 2xy + 0 + y = y(2x + 1), \quad \nabla \cdot \mathbf{F}|_{\mathbf{p}} = 3.$$

Curl, from (0.7.33):

$$\begin{aligned} \nabla \times \mathbf{F} &= (\partial_y(yz) - \partial_z(3x - z^2), \partial_z(x^2y) - \partial_x(yz), \partial_x(3x - z^2) - \partial_y(x^2y)), \\ &= (z + 2z, 0 - 0, 3 - x^2) = (3z, 0, 3 - x^2), \quad \omega|_{\mathbf{p}} = (3, 0, 2). \end{aligned}$$

The split. By (0.7.34), at $\mathbf{p}$:

$$S = \frac{1}{2}(J + J^T) = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 0 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 1 \end{pmatrix}, \quad A = \frac{1}{2}(J - J^T) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\frac{3}{2} \\ 0 & \frac{3}{2} & 0 \end{pmatrix}.$$

The checks. $\text{tr } S = 2 + 0 + 1 = 3 = \nabla \cdot \mathbf{F} \checkmark$, and $\text{tr } A = 0 \checkmark$, confirming (0.7.35): the whole divergence lives in the symmetric part. Now compare A with the prediction (0.7.37) using $\omega = (3, 0, 2)$:

$$\frac{1}{2} \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -2 & 0 \\ 2 & 0 & -3 \\ 0 & 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\frac{3}{2} \\ 0 & \frac{3}{2} & 0 \end{pmatrix}.$$

Identical, entry by entry. $\checkmark$ And the cross-product form (0.7.38), tested on $\mathbf{h} = (1, 2, 3)$:

$$\begin{aligned} A\mathbf{h} &= (-2, -\frac{7}{2}, 3), \\ \frac{1}{2}\omega \times \mathbf{h} &= \frac{1}{2}(3, 0, 2) \times (1, 2, 3) = \frac{1}{2}(-4, -7, 6) = (-2, -\frac{7}{2}, 3). \checkmark \end{aligned}$$

What the numbers say physically. Treat $\mathbf{F}$ as a velocity field. A small blob of tracer sitting at $(1, 1, 1)$ is doing three things at once. Its volume is growing at a fractional rate of 3 per unit time, by (0.7.26). It is spinning rigidly at angular velocity $\frac{1}{2}\omega = (1.5, 0, 1)$, an axis lying in the xz -plane, with angular speed $\frac{1}{2}\sqrt{3^2 + 2^2} = \frac{1}{2}\sqrt{13} \approx 1.80$ radians per unit time. And it is being strained according to S , whose eigenvectors (Chapter 0.5) are the principal axes of the distortion and whose eigenvalues are the stretch rates along them — necessarily summing to 3, since $\text{tr } S$ is the divergence. Every number in the Jacobian has been accounted for: three in the trace, three in the curl, and the remaining five in the traceless symmetric part that neither operator sees.

9 • Your turn

PROBLEM 1 — VERIFY THE DIVERGENCE THEOREM, BOTH SIDES

(a) For $\mathbf{F} = (xy, y^2, z)$ and V the unit cube $0 \leq x, y, z \leq 1$, compute $\int_V \nabla \cdot \mathbf{F} \, dV$ and $\oint_{\partial V} \mathbf{F} \cdot d\mathbf{A}$ separately — all six faces — and check that they agree. (b) For $\mathbf{F} = \mathbf{r} = (x, y, z)$ and V a ball of radius R , do the same, and read off a formula for the volume of any region in terms of a surface integral.

- Solution

(a) Volume side. $\nabla \cdot \mathbf{F} = \partial_x(xy) + \partial_y(y^2) + \partial_z(z) = y + 2y + 1 = 3y + 1$. Over the unit cube the x and z integrations give factors of 1:

$$\int_V (3y + 1) \, dV = \int_0^1 (3y + 1) \, dy = \frac{3}{2} + 1 = \frac{5}{2}.$$

Surface side. Six faces, each with area element 1 and outward normal $\pm$ a coordinate direction. Only the component of $\mathbf{F}$ along that normal contributes.

FACE	$\hat{\mathbf{n}}$	$\mathbf{F} \cdot \hat{\mathbf{n}}$ THERE	INTEGRAL
$x = 1$	$+\hat{\mathbf{x}}$	$xy = y$	$\int_0^1 \int_0^1 y \, dy \, dz = \frac{1}{2}$
$x = 0$	$-\hat{\mathbf{x}}$	$-xy = 0$	0
$y = 1$	$+\hat{\mathbf{y}}$	$y^2 = 1$	1
$y = 0$	$-\hat{\mathbf{y}}$	$-y^2 = 0$	0
$z = 1$	$+\hat{\mathbf{z}}$	$z = 1$	1
$z = 0$	$-\hat{\mathbf{z}}$	$-z = 0$	0

Total flux $= \frac{1}{2} + 1 + 1 = \frac{5}{2}$. ✓ Equal to the volume integral, as (0.7.44) demands.

Worth noticing *why* three faces contributed nothing: on each of them the relevant field component vanishes. The theorem does not care; it only asks that the totals match.

(b) $\nabla \cdot \mathbf{r} = 1 + 1 + 1 = 3$, so the volume side is $3 \cdot \frac{4}{3}\pi R^3 = 4\pi R^3$. On the sphere $\hat{\mathbf{n}} = \hat{\mathbf{r}}$ and $\mathbf{r} \cdot \hat{\mathbf{r}} = r = R$, a constant, so the flux is $R \cdot 4\pi R^2 = 4\pi R^3$. ✓

Since $\nabla \cdot \mathbf{r} = 3$ for *any* region, the divergence theorem gives a general formula for volume as a surface integral:

$$\text{vol}(V) = \frac{1}{3} \oint_{\partial V} \mathbf{r} \cdot d\mathbf{A}.$$

That is how a CAD package computes the volume of a mesh: it never fills the interior, it just sums a quantity over the triangles of the surface. The divergence theorem is why that works.

PROBLEM 2 — THE FIELD WITH NO POTENTIAL

For $\mathbf{F} = (-y, x)/(x^2 + y^2)$ on the punctured plane: (a) confirm $(\nabla \times \mathbf{F})_z = 0$ everywhere it is defined; (b) compute $\oint \mathbf{F} \cdot d\mathbf{r}$ around the circle of radius R centred at the origin, and around the circle of radius 1 centred at $(3, 0)$; (c) explain, in a paragraph, what the two results together imply about the existence of a potential, and identify precisely which hypothesis of which theorem fails.

– Solution

(a) Done in (0.7.10): both $\partial_x F^y$ and $\partial_y F^x$ equal $(y^2 - x^2)/(x^2 + y^2)^2$, so their difference vanishes at every point of the punctured plane.

(b) Around the origin, (0.7.12) gives 2π for every R ; the radius cancels because $|\mathbf{F}| = 1/R$ while the circumference is $2\pi R$.

For the circle of radius 1 about $(3, 0)$: this loop does *not* enclose the origin, so it bounds a disc lying entirely inside the domain, and Stokes' theorem (0.7.41) applies to give $\oint \mathbf{F} \cdot d\mathbf{r} = \int (\nabla \times \mathbf{F}) \cdot d\mathbf{A} = 0$ by part (a). Zero, with no integration needed. (Direct numerical evaluation of the parametrised integral confirms it to machine precision.)

Two loops in the same field, one giving 2π and one giving 0. The difference is not the field — it is where the loops are.

(c) By §2.3, a potential exists on a region if and only if every closed loop in that region has zero circulation. The loop around the origin has circulation $2\pi \neq 0$, so **no single-valued potential exists on the punctured plane** — even though the field is curl-free at every single point of it. Locally the potential exists and equals the polar angle θ ; globally θ is not a function, since going once round increases it by 2π , and that increment is exactly the circulation we computed.

The failing hypothesis is precisely identified. The implication "curl-free $\Rightarrow$ conservative" is proved by taking a loop, spanning it with a surface, and applying Stokes' theorem — which requires the surface to lie inside the region where $\mathbf{F}$ is defined. Any surface spanning a loop around the origin must pass through the origin, where $\mathbf{F}$ does not exist. So Stokes' theorem is not available, and the implication is not owed to us. On a simply connected region — a disc, or the right half-plane, or anything with no hole — every loop *does* bound such a surface and the implication is restored. Note what this means practically: the local test (D) can never detect the obstruction, because the obstruction is not local. You have to know the shape of the domain.

PROBLEM 3 — HOW LONG DOES DIFFUSION TAKE?

(a) From the diffusion equation $\frac{\partial C}{\partial t} = D\nabla^2 C$, use dimensional analysis (Chapter 0.3) to show that the only length scale available after a time t is $\sqrt{Dt}$, and say what dimensional analysis cannot tell you here. (b) For a molecule with $D \approx 10^{-6} \text{ cm}^2/\text{s}$, estimate the time to diffuse 100 μm , then 1 mm, then 1 cm. (c) Interpret: why is 100 μm a number that keeps appearing in histology?

– Solution

(a) Read the units off the equation itself. The left side has dimensions $[C]/[T]$; the right side has $[D][C]/[L]^2$. Equating,

$$[D] = \frac{[L]^2}{[T]}.$$

The problem supplies exactly two quantities with dimensions, D and t , and one combination of them has dimensions of length: $\sqrt{Dt}$, uniquely, since $D^a t^b$ has dimensions $L^{2a} T^{b-a}$, and $L^1 T^0$ forces $a = \frac{1}{2}$, $b = \frac{1}{2}$. Hence $\ell = c\sqrt{Dt}$ for some pure number c .

What dimensional analysis cannot give is c — and c depends on what you mean by "the distance diffused". Solving the equation gives a Gaussian profile with $\langle x^2 \rangle = 2Dt$ in one dimension and $\langle r^2 \rangle = 6Dt$ in three, so c is $\sqrt{2}$ or $\sqrt{6}$ or 1 depending on the question. That is a factor of two or so in t , which for an order-of-magnitude question does not matter and for a quantitative one does. Same lesson as Chapter 0.3's pendulum: the scaling is free, the constant is not.

(b) Using $t = \ell^2/D$ with $D = 10^{-6} \text{ cm}^2/\text{s}$ and remembering $100 \mu\text{m} = 10^{-2} \text{ cm}$:

$$t = \frac{(10^{-2} \text{ cm})^2}{10^{-6} \text{ cm}^2/\text{s}} = \frac{10^{-4}}{10^{-6}} \text{ s} = 100 \text{ s}.$$

Then 1 mm = 10^{-1} cm gives $t = 10^{-2}/10^{-6} = 10^4 \text{ s} \approx 2.8 \text{ hours}$, and 1 cm gives $t = 1/10^{-6} = 10^6 \text{ s} \approx 11.6 \text{ days}$.

(c) The quadratic is the whole story: ten times the distance costs a hundred times the time. Under two minutes to cross 100 μm is entirely compatible with a cell's metabolic timescale; twelve days to cross a centimetre is not compatible with anything. So diffusion is a viable transport mechanism up to roughly a hundred microns and useless beyond it, and every tissue is built to respect that boundary — capillaries spaced tens of microns apart, alveolar walls under a micron thick, and tumour cords with a viable rim of about 100–150 μm and a necrotic centre. Anything that must travel further than the diffusion limit is moved by bulk flow instead. The number in the histology is the number in (0.7.63).

Worth noting what this implies about therapy. If a drug must penetrate 200 μm from the nearest vessel rather than 100, it does not take twice as long — it takes four times as long, during which it is being

cleared. Penetration depth at fixed clearance goes like the square root of everything, which is why it is so stubbornly hard to improve.

PROBLEM 4 — THE IDENTITY THAT MAKES ELECTROMAGNETISM WORK

(a) Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for any twice-continuously-differentiable $\mathbf{A}$, stating exactly which theorem you use. (b) Take the divergence of Faraday's law $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$ and say what it tells you about $\nabla \cdot \mathbf{B}$. (c) Take the divergence of the Ampère–Maxwell law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$, use Gauss's law $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, and interpret the result. (d) What would break if the identity in (a) were false?

– Solution

(a) Expand as in (0.7.55):

$$\nabla \cdot (\nabla \times \mathbf{A}) = \partial_x \partial_y A^z - \partial_x \partial_z A^y + \partial_y \partial_z A^x - \partial_y \partial_x A^z + \partial_z \partial_x A^y - \partial_z \partial_y A^x.$$

The first and fourth terms cancel, as do the second and fifth, and the third and sixth — in each case because the two differentiations are in the opposite order on the same component. The theorem used is **Clairaut's** (Chapter 0.6 §6.1), whose hypothesis is that the second partials are continuous; that is why the problem said twice *continuously* differentiable. Nothing else is used, and in particular no property of $\mathbf{A}$ beyond smoothness.

(b) Taking the divergence of Faraday's law, the left side vanishes by (a), and the right side is $-\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B})$ since space and time derivatives commute. So

$$\frac{\partial}{\partial t}(\nabla \cdot \mathbf{B}) = 0.$$

This is a genuinely nice result: $\nabla \cdot \mathbf{B}$ cannot change in time. So "there are no magnetic monopoles" is not an independent dynamical law that has to hold at every instant — it is an *initial condition*. Set $\nabla \cdot \mathbf{B} = 0$ once and Faraday's law keeps it zero forever. Maxwell's equations are internally consistent about this because of (a).

(c) Same move on Ampère–Maxwell. The left side vanishes by (a), so

$$0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \cdot \mathbf{E}) = \mu_0 \left(\nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t} \right),$$

using Gauss's law in the last step. Dividing by μ_0 :

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

That is the continuity equation (0.7.50). **Charge conservation is not an extra assumption in electromagnetism — it is forced by the structure of Maxwell's equations, and the thing doing the forcing is the identity in (a).**

Historically this is how the displacement current was found. Without the $\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$ term, the same computation would give $\nabla \cdot \mathbf{J} = 0$, which is false whenever charge accumulates anywhere — a capacitor charging, for instance. Maxwell added the term precisely to repair the inconsistency, and the repaired

equations then predicted electromagnetic waves. A term added to make a mathematical identity come out right turned out to be light.

(d) Three things break, in increasing order of severity. First, $\mathbf{B} = \nabla \times \mathbf{A}$ would no longer automatically satisfy $\nabla \cdot \mathbf{B} = 0$, so the vector potential could not be used to represent a physical magnetic field — and with it goes the entire Lagrangian and Hamiltonian formulation of electromagnetism, the Aharonov–Bohm effect, and the gauge principle of Chapter 6.3, all of which are written in terms of $\mathbf{A}$ rather than $\mathbf{B}$. Second, by (b) $\nabla \cdot \mathbf{B}$ would be free to develop in time, so monopoles could appear from nothing. Third, and worst, by (c) Maxwell's equations would be flatly inconsistent with charge conservation: the two curl equations would impose a condition on ρ and $\mathbf{J}$ that real charges do not satisfy, and the system would have no solutions for generic sources.

All of that rests on the commutativity of mixed partial derivatives. It is worth appreciating how much weight Clairaut's theorem is carrying.

You now have the calculus of fields. **Line integrals**, with the gradient theorem showing that a conservative field's line integral is the Fundamental Theorem in disguise — and the equivalence chain (A)–(D) with its one broken link, which is broken by *topology* and not by analysis. **Divergence**, defined coordinate-free as flux per unit volume, derived into $\partial_i F^i$ by a box, and then recognised as the *trace of the Jacobian*, which by Chapter 0.4 makes it the fractional rate of volume change of a flowing blob. **Curl**, defined as circulation per unit area, derived by a rectangle, and recognised as the *antisymmetric part of the Jacobian* — twice the local angular velocity — with the count $\frac{1}{2}n(n-1) = n$ explaining why it can pretend to be a vector in three dimensions and nowhere else. **One theorem** in four costumes, all four proved by the same interior-cancellation argument that Chapter 0.2 used on an interval. The **continuity equation**, which is what a conservation law is. And the two **second-derivative identities**, which between them create the vector potential and its gauge freedom, plus the **Laplacian**, read as the gap between a field and its own local average.

Where this gets spent.

- **Line integrals and potentials** → Chapter 2.6 and Chapter 6.3, where A^μ becomes fundamental; and the Aharonov–Bohm effect, which is §2.4 measured in a laboratory. The lesson that a curl-free field need not be a gradient is the first topological fact in this book and it recurs in Chapters 6.5 and 7.7.
- **Divergence as trace** → Chapter 1.3, where Liouville's theorem is (0.7.26) plus Clairaut and takes one line; and Chapter 5.5, where the same determinant reappears as a Jacobian in a functional integral.
- **Curl as the antisymmetric part** → Chapter 2.6, where four dimensions leave the antisymmetric object no vector to hide in and it becomes $F_{\mu\nu}$; Chapter 6.1, where antisymmetric matrices become the Lie algebra of the rotation group.
- **The unified Stokes theorem** → Chapter 3.5, which builds differential forms and proves $\int_M d\omega = \oint_{\partial M} \omega$ once and for all; and everywhere afterwards, because conservation laws, Gauss's law, and the Bianchi identities are all this one statement.
- **The continuity equation** → Chapter 4.4 (probability current), Chapters 1.4 and 5.2 (Noether currents), Chapter 3.6 ($\nabla_\mu T^{\mu\nu} = 0$). If you learn to recognise $\partial_\mu J^\mu = 0$ on sight you will find it in every part of this book.
- $d^2 = 0$ → Chapter 3.5, where the two identities of §7 become one; and Chapter 3.6, where the same statement (as the Bianchi identity) is what forces energy-momentum conservation in general relativity rather than merely permitting it.
- **The Laplacian and Poisson's equation** → Chapter 0.9 (where ∇^2 meets the delta function and the 4π is finally accounted for), Chapter 4.4 (the kinetic term of the Schrödinger equation is a Laplacian), and Chapter 3.6, where the Einstein field equations collapse to $\nabla^2 \Phi = 4\pi G\rho$ in the Newtonian limit.

One sentence to carry above all others. Chapter 0.2 said: integrate a derivative over a region and the interior cancels, leaving the boundary. This chapter said it three more times in higher dimensions and

gave the result three different names. Chapter 3.5 will say it once. Everything in between — Gauss's law, Stokes' theorem, conservation of charge, conservation of energy-momentum — is bookkeeping on that one idea.